

Systematic Innovation



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In this month's issue:

Article – Defining **EQ** (Forty Years Too Late)

Article – AI Trajectories – Winters, Disruptions & Emotional Intelligence

Not So Funny – Asimov's Laws

Patent of the Month – Small-Scale Robots For Biofilm Eradication

Best of The Month – The Lady's Handbook For Her Mysterious Illness

Wow In Music – Kashmir

Investments – Biodegradable Plastic

Conference Report – TRIZCON

Generational Cycles – From Duty to Self: How Generational Shifts in Wealth Are Redefining Legacy and Philanthropy

Biology – Human Ear

Short Thort

News

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Readers' comments and inputs are always welcome.

Send them to darrell.mann@systematic-innovation.com

Defining EQ (Forty Years Too Late)

Forty years ago this year, the phrase 'emotional intelligence' was first used. Happy Birthday, and a shout out to PhD researcher, Wayne Payne's 1985 doctoral dissertation "A Study of Emotion: Developing Emotional Intelligence". Wayne's spark took a few years before it caught light – 1989 saw Stanley Greenspan used similar ideas in developmental psychology; 1990, Peter Salovey and John Mayer formally defined emotional intelligence in academic literature – but the fire exploded into the zeitgeist in 1995 when Daniel Goleman popularised the concept in his bestselling book Emotional Intelligence. Goleman can also be credited with giving the world the term 'EQ' as an accessible shorthand that paired with the already ubiquitous, IQ.

The short history of both emotional intelligence and EQ is perhaps a good indicator of the maturity – or lack thereof – of both concepts. From where we stand – and please bear in mind this is an engineer talking, so not the sort of person typically well endowed with emotional intelligence – what Goleman achieved is a first ascent onto a collective Mount Stupid. For almost everyone. We all, I believe, think we know what EQ is. And I believe we all also think we know it when we see it. When we started the research for the 1%ers book, we kind of took it as read that 'high EQ' was one of the key characteristics of the rare person capable of getting new things done. The further we get into the world of generative AI, the more EQ becomes the thing that keeps the humans ahead of the getting new things done game. EQ becomes much more important than IQ, because the latter is the sort of thing that logic-based AIs can already beat most humans at. When the Artificial General Intelligence barrier is broken (2027 seems to be the latest prediction), what will have been broken is any hope of humans beating the AI at any kind of IQ-oriented task.

The importance of 'high EQ' has been further evidenced by the shift in the type of work we find ourselves doing when working with clients on innovation projects: twenty-five years ago, pretty much everything we did was supply the 'someone, somewhere already solved your problem' IQ parts of the innovation equation. These days, pretty much everything has evolved into dealing with all the 'people issues' surrounding the (easy) IQ bits – consumers that tell you one thing when they mean something completely different; team members that tell you they're on-board, but aren't; committed sponsors that prove themselves to be highly non-committal when things get tricky elsewhere in the business; competitors that decide a lengthy social media flame-war is the best way of out-competing us; regulators that take it upon themselves to fail audits that we sailed through with flying colours every year for the last decade. The list of human foibles hazarding the successful completion of getting new things done work is apparently endless.

And yet, the 1%er somehow manages to deal with all of them and still get things delivered on target, on budget and on time. We know these people when we see them. But, the deeper we look, the more we realise, there was really no effective way to measure what 'good' or 'high' actually meant. And that's a problem. A very big problem if part of the goal of producing the book is to help create more 1%ers. If we can't measure it, how can we hope to improve it? So goes the formal logic.

Let's have a quick look at the EQ-measurement state of the art to see where the sticking points and opportunities to do better might be. There seem to be three basic types of assessment tool on the market:

1. Ability-based tests (objective tasks): Example: Mayer–Salovey–Caruso Emotional Intelligence Test (MSCEIT) where people are asked to solve problems involving emotions: identifying emotions in facial expressions or artwork, predicting how emotions change in scenarios, choosing the best way to regulate emotions, with scoring then based on consensus (how most people rate an answer) or expert judgment.
2. Self-report questionnaires (personality-style inventories): Example: EQ-i 2.0 by Bar-On or Schutte Emotional Intelligence Scale where people rate themselves on statements like “I handle stressful situations well” or “I can sense how others feel.”
3. 360-degree or mixed models (behavioural ratings): Example: Emotional Competence Inventory (Goleman model) where colleagues, bosses, or clients rate the person’s emotional skills in action – empathy, self-control, social influence, etc.

The third type is probably the most meaningful, albeit being very context-dependent – a person may look high-EQ at work and low-EQ at home – it does at least acknowledge the important question of application. Same thing with IQ – knowing where a person is on the scale is of little value in determining what they can or cannot ‘do’. The highest IQ (or EQ) person on the planet is of little use to anyone if their capabilities aren’t used to deliver some kind of useful outcome.

The other two types of test both completely fail in this regard – the ability-based tests measure knowledge about emotions, not whether a person can actually apply it under stress, and the self-report questionnaires, well, what can we politely say, are at best highly vulnerable to social desirability bias (people think they’re empathetic, so they check the box). Neither test real-world behaviour. And arguably, given Heraclitus’ proposition that we can never step in the same river twice, neither does the third.

The overall upshot being, we collectively don’t know how to measure EQ, and, worse, we don’t really know *why* we’re measuring it.

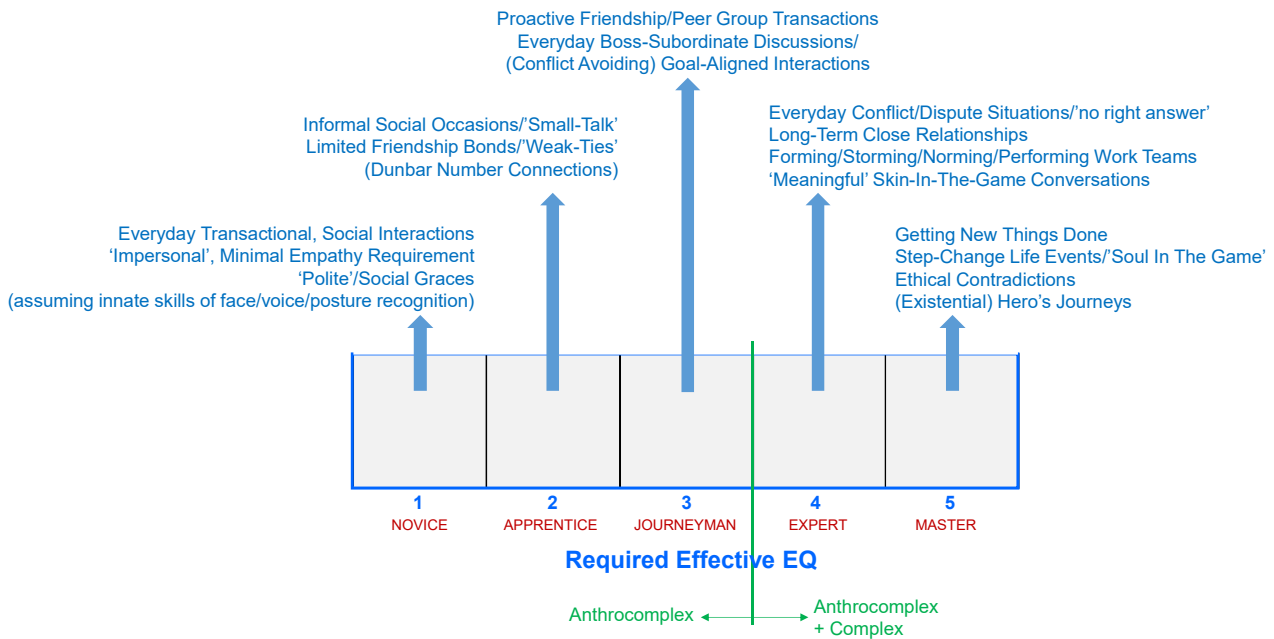
Given that the ‘why’ of a 1%er is all about getting new things done, it felt like we needed to re-think the whole EQ story if we had any intention of being able to create more of these apparently (according to our instincts) ‘high EQ’ people.

We started that journey by analysing the spectrum of EQ ‘jobs to be done’ and then dividing that spectrum up into discontinuously different types of EQ capability. In effect s-curves. Such that one EQ s-curve would allow us to get certain types of job done – successfully dealing with a check-out assistant in the supermarket, for example, or negotiating a pay-rise with the boss – but other jobs would require a step-change shift in capability if we are to achieve a successful outcome. We ended up with five stages as shown in the chart over the page.

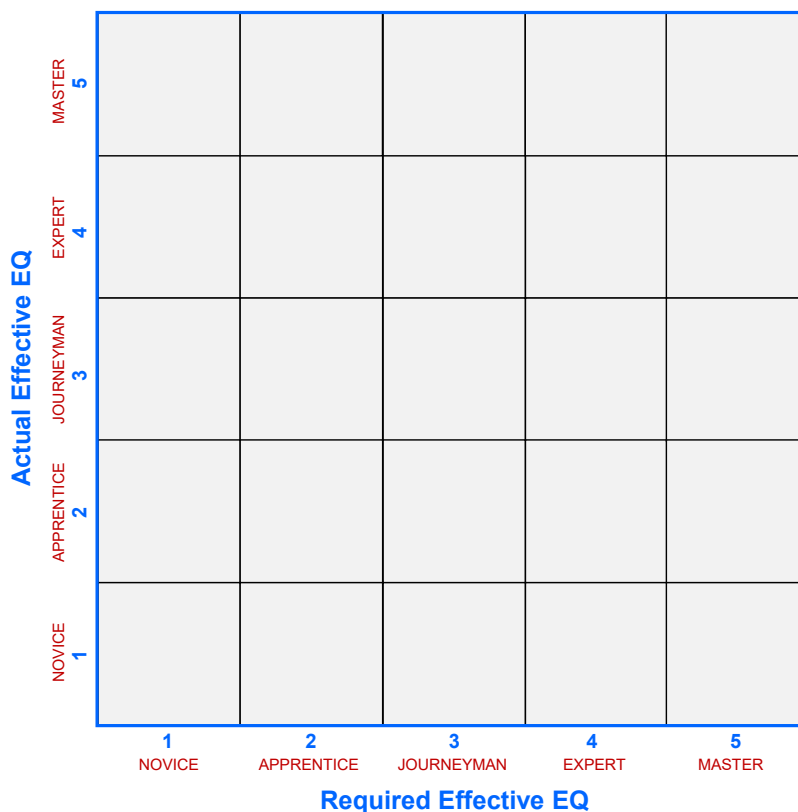
Over on the right-hand side of the chart – ‘Level 5 EQ’ – demands the kind of mastery that 1%ers somehow manage to have acquired. The sort of Mastery required to successfully follow the Hero’s Journey, deal with step-change life events or anything else that might get classed as existential.

Effective EQ at any of the five levels we’ve identified starts with the amazing (‘innate’?) human abilities to recognise the meaning of facial expressions, micro-expressions, tones of voice and posture. Our huge brains have evolved to be good at dealing with ‘anthrocomplexity’. All of these ‘innate’ skills hum away in the background across all five stages of the requirement spectrum, especially the first three. By the time we encounter ‘Expert’, Level 4 requirements, we’ve crossed the other complexity boundary and find

ourselves in the realm where our anthrocomplexity capabilities need to be combined with an understanding of complex adaptive systems if the requirements of the task at hand are to be satisfied.

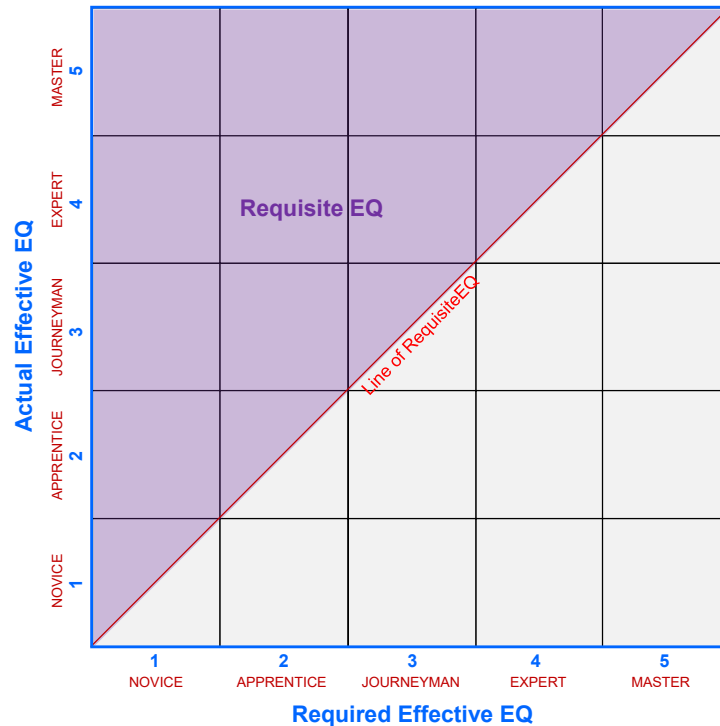


Having identified this spectrum of required-EQ Levels, it was then easy to add an orthogonal dimension describing an equivalent five-stage spectrum of 'actual' EQ Levels:

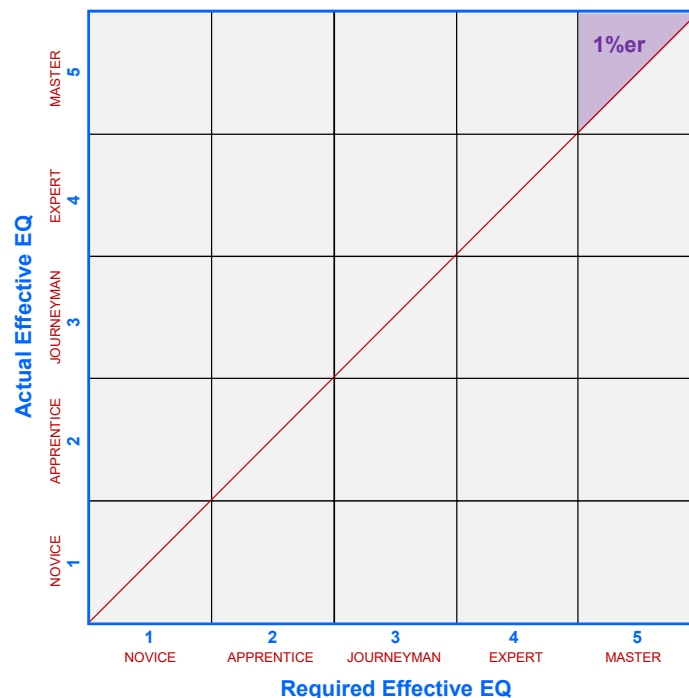


This now offers a map of the world that, assuming we can work out what 'required' and 'actual' EQs are, enables us to examine how the world works. Or, as often transpires, doesn't.

The first – hopefully obvious – addition to the map is a diagonal line we can draw from the bottom-left to the top-right corner. Such a line denotes situations where the level of actual EQ is the same as the EQ required to successfully achieve a desired outcome. We might think of this diagonal line as a line of 'Requisite EQ' and that, anywhere on the map above that line is a good place to be because our *actual* EQ is greater than that *required*:



We can also see that, hypothetical as it might be at this point in the discussion, the small triangle above the RequisiteEQ line in the Master-Master square in the top-right hand corner of the map is where a 1%er sits – they are required to get new things done (i.e. a Level 5 Requirement), and have the requisite (or more) actual EQ to deal with whatever the world decides to throw at them (i.e. are above the RequisiteEQ line):



If you like, that 1%er triangle acts as a kind of a compass heading for the aspiring 1%er.

For some reason, the twinned-worlds of psychology and psychiatry seem somehow reluctant to acknowledge the idea of directionality or the concept of a direction that somehow implies that humanity itself has any kind of trajectory, especially if such directions in any way imply 'getting better'. I'm not sure whether this comes as part of the science-envy in the two professions (many scientists get very offended when we start even suggesting that life has any kind of 'purpose' – see McGilchrist for a long discussion on the flaws in this thinking) or some kind of reaction against the 'Übermensch' concept in the philosophy of Friedrich Nietzsche, first introduced in his 1883 book *Thus Spoke Zarathustra* and often translated as "overman" or "superman". It represented an ideal for humanity to strive for, a self-overcoming being that transcends conventional moral and societal limitations, creating its own values and affirming life fully. Nazi Germany's 'master-race' delusions probably poisoned that well.

Let us therefore be clear that a 1%er is no kind of 'Übermensch'. They are no better or worse than any other human on the planet. Some jobs require certain skills, and when it comes to getting new things done, it just happens to be the 1%er capability set that is best qualified to get them done. Oh, and our omni-crisis world seems to need more of them right now. That doesn't mean that we'd necessarily want to invite them to our next party. Or bowling night.

The idea of directionality, meanwhile, is an important one. Just because Nietzsche's thinking was corrupted doesn't mean the basic concept was necessarily wrong. When it comes to creating successful innovation attempts, for example, all the TRIZ/SI research clearly shows the existence of a crystal clear and universal direction of success. A direction defined by increasing value – aka 'more benefits, less costs, less harms' – and that ultimately that means heading towards an 'Ideal Final Result'.

When it comes to determining directionality as far as human emotions are concerned, McGilchrist (again) makes the best argument yet that, over time, truth, beauty and goodness increase. I can buy into all three, but at this point in history I'm not going to make any of my arguments rely on convincing others that they're right. Never mind, to push McGilchrist's argument further, that they are the *only* three.

Far easier is to acknowledge Nassim Taleb's 'AntiFragile' as the safest definition of a meaningful 'getting better' compass heading for humans to follow. Ask anyone whether they'd prefer to be fragile or antifragile and an immediate consensus is produced: no-one wants to be fragile. Everyone would like to be heading in the direction of increasing antifragility. No evidence and no quantification required, we somehow instinctively *know* that antifragile is a better place to be. Perhaps because, for most of human history, we have evolved many highly antifragile traits both physically and emotionally. We are antifragile. Or rather, we were. Through the last couple of hundred years, and the last eighty years in particular, we've somehow allowed ourselves to be nudged in the opposite direction.

Fortunately, we don't need to dig any further in that direction as far as the discussion here is concerned. Being in the 1%er triangle at the top-right hand corner of the EQ Map is a good place to be, if only, to repeat, because the world needs more people that can get new stuff done. And if they happen to be antifragile, understand increasing value, truth, beauty and goodness as well, so be it.

What directionality, meanwhile, does for us is help work out ways to make meaningful measurements of EQ. Measurements that escape the trap of 'high EQ' in *theory* versus a more useful measure of *effectiveness in practice*.

As is often the case, we learn more and faster when we examine extreme situations. Especially ones where the range of outcomes obtained is variable to the point of randomness... let's take a dive into the world of clinical psychology, psychiatry and, trigger word warning, 'therapy'.

The most likely times in life when a patient is going to seek out help from a member of one of these professions is after some kind of traumatic life event. In other words, a situation requiring Level 5 EQ 'Mastery' skills. A situation that is therefore both anthrocomplex and complex.

A couple of decades ago, feeling a little lost myself, I decided to sign up for a formal psychology qualification. I lasted about three months. Everything started okay. Then I started asking awkward questions about 'efficacy' and 'effectiveness'. The first thing I noticed was how much these questions annoyed my lecturers. As if I was somehow missing the point of what they were trying to lecture me about. Nevertheless, I pressed on until we reached a point in the clinical part of the curriculum and the emergence of some actual outcome data. The likelihood of a Jungian psychologist 'curing' a patient, we were informed, was 'around 30%'. For Freudians the figure was about the same. And for those patients that never received any treatment, it was also about the same. Now, ignoring the slightly awkward question of what does 'cure' actually mean, for a moment, if the outcome is the same if a patient does or doesn't seek help that would seem to apply to me that there is something not right with the system.

The first thing not right about it is the fairly big elephant in the room that none of the lecturers I encountered had even the vaguest inkling of what I was talking about when I asked them about systems. They had no conception of what a system was, or why it might be important to their work with a patient.

This, I believe, ought to be a pretty big red flag. If a psychologist is going to 'cure' a patient, it requires a system. The reason the patient is a patient in the first place is because the systems they found themselves part of no longer did what they had been doing. Systems 101: the purpose of a system is what it does. POSIWID. If the patient is so anxiety-ridden they can't get up off the bathroom floor some mornings, its not because that's what they 'decided' to do, it's because of what the systems they're a part of are doing to them.

The potential irony at this point, of course, is that 'it's not your fault, it's the system' sounds strikingly similar to the secret of both Jung and Freud's near-universal spread across the whole of the psychology domain. Their message to patients in effect was and still is, 'it's not your fault'. It was your parents. Or some pre-adolescent trauma. Or something from a previous life or dream that made you like this.

I say 'near universal' because Alfred Adler, a contemporary of Jung and Freud, took the rather different view of saying to patients, 'it is your fault'. Or, if that sounds a little harsh, 'the past is irrelevant because there's nothing you can do about it, you can only control what you can control, and that means you and what you choose to do next'. Adler, in other words, was promoting antifragility. That everyone has heard of Jung and Freud, and almost no-one has heard of Adler still, is a testament to market forces, and the general human preference for comforting lies over uncomfortable truths.

I'm not sure that Adler understood systems either, but somehow he understood them instinctively enough to tell patients that the only thing they had control over was the perspective they took on what they saw happening around them. The patient doesn't decide the outcome, the system decides the outcome. All the patient can do is control how they perceive that system.

A better overall working hypothesis – better than Jung, Freud or Adler – would be that if a therapist actually understood complex systems, they might actually be at least somewhat equipped to help their patients to work on their surrounding systems if they wanted to achieve different outcomes to the ones being produced currently. Nowhere in the psychology/psychiatry/therapy academic literature anywhere is there any evidence of an understanding of complex adaptive systems. Which means lots of compiling of symptoms and circumstances gets done, but no attempt to map the relations *between* any of them is sought. And therefore half the picture is eliminated.

If the similarities here between dysfunction in the therapy space and the innovation space are beginning to become apparent, let's complete the set by bringing the word 'contradiction' into the discussion. The 3Cs of a successful innovation attempt – Compass, Complexity, Contradiction – start to look strikingly similar to what a 'cured' patient might look like. Every Hero's Journey comes with its Ordeal (contradiction) – the impossible problem that somehow, magically and unexpectedly, the Hero manages to conjure up at the moment when all seems lost.

This feels like another of those Taleb 'Antifragile' evidence moments. Do I need any evidence to convince anyone that solving contradictions is better than making trade-offs and compromises?

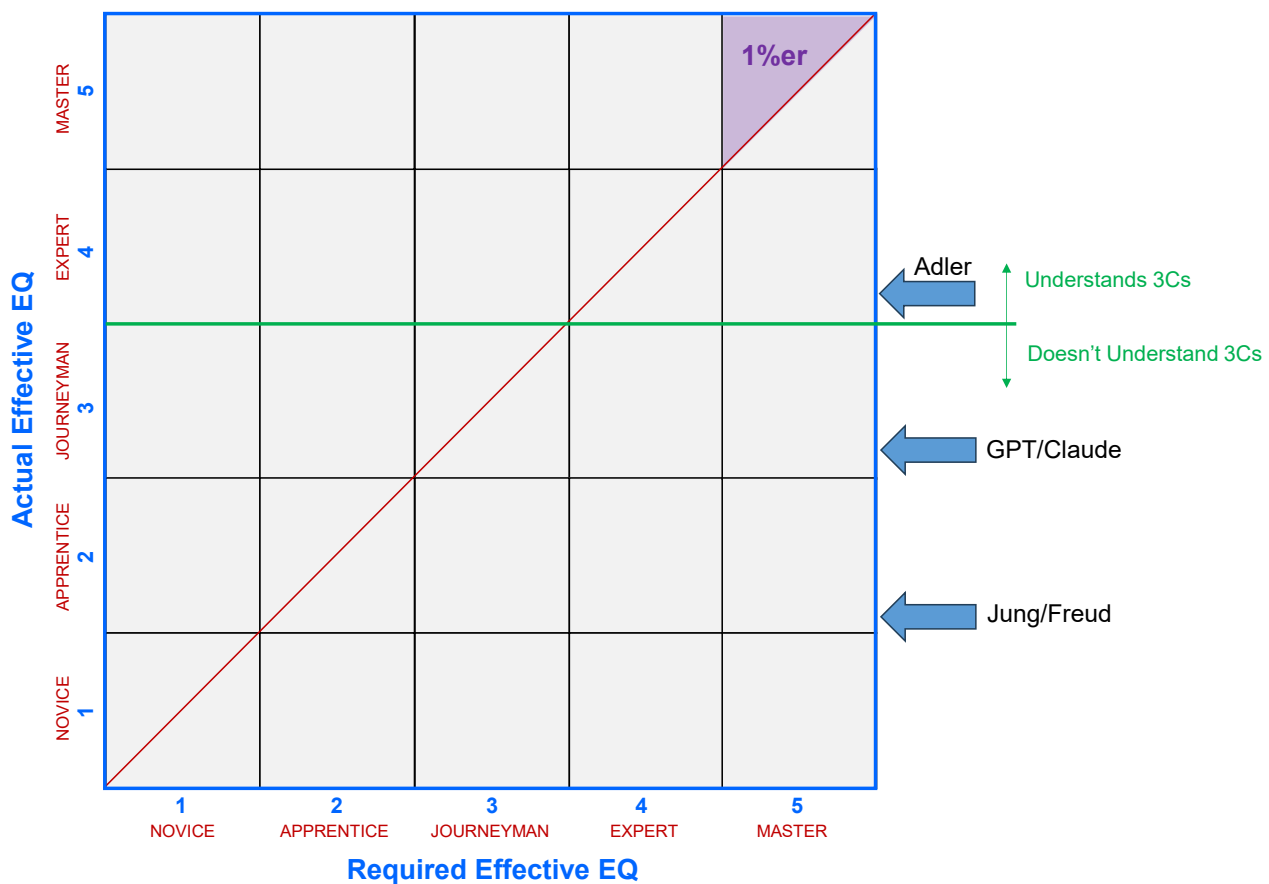
Perhaps I've spent too many years proving that it's possible to solve contradictions to have any doubt that solving them is better than not solving them. I got to that point, too, thanks to thirty years of decoding innovation success. I've not had the same amount of time to try and decode 'curing' patients in the world of therapy, but I do have the advantage of AI tools that know how to go and read unstructured narrative – and pretty much the whole of the therapy corpus of knowledge is in the form of very unstructured patient notes – to find contradictions, whether they got resolved and how they were resolved. And, having now had the opportunity to run large corpuses of patient notes through that software, I can safely conclude that therapists don't look for contradictions, ignore them if they accidentally reveal them, and therefore preclude breakthrough thinking when it comes to recommending any kind of advice to patients. The contradiction solving cupboard is bare, psychology-wise.

I think this is a problem.

I think if I was looking for a therapist to get me through a difficult time, I'd want one who understood the 3Cs.

I think a 1%er knows those 3Cs – explicitly or implicitly – as far as the new things they're responsible for getting done are concerned. I similarly think that they also – explicitly or implicitly – recognise it is the same 3Cs that will determine whether they deal with the 'people issues' effectively.

And if I take that hypothesis back to the EQ Map and the Level 5 challenge, I think the current therapy state-of-the-art looks something like this:



Outside the 1%er world, I think Adler and the slowly growing list of acolytes get closest to achieving the Requisite Level of actual EQ required to deal with Level 5 requirements. He at least understood one of the 3Cs, and possibly enough of the second C to enable patients and followers to successfully deal with some Level of complexity.

Jungians and Freudians, on the other hand, because they took the financially rewarding and hence industry-building direction of making patients more fragile have ended up falling victim to the Basic Laws Of Human Stupidity. Having good intentions – and I have to say that every psychologist, psychiatrist and therapist I've ever met admirably ticks that box – is not enough. Good intentions plus inadequate knowledge begets bad outcomes.

Meanwhile, there's a reason that increasing numbers of therapy-seekers are finding themselves talking to a generative-AI. For a start it's cheaper. And, probably because the AI seems to have built-in Marcus Aurelius and the Stoics (good name for a band) LLMs, the advice they will return can easily sound as plausible as any than that which an actual therapist might propose.

Crucially, however, whether it's a Jungian therapist, or Claude or Adler, if the task at hand requires Level 5 EQ, none offer the requisite EQ to get to the desired outcome. No-one in the therapy system – patient included – knows how to ask the right questions, or interpret the answers in a meaningful way. The therapy state of the art for Level 5 requirement situation is below the RequisiteEQ line and is therefore essentially a crap-shoot.

That's not a good thing. But, what's good about it is, the more the capability vacuum becomes visible, the greater the likelihood that someone or something comes along to fill that vacuum. Something requisite. And very likely someone 1%er shaped.

Again, being a 1%er doesn't make you any better or any worse than any other human being on the planet. It merely equips you to deal with a broader range of situations. Situations that an AI will likely never match.

That said, we're all still on Mount Stupid when it comes to EQ. By way of one small step in the ascent beyond Mount Stupid, interested readers may be interested in trying out our new online 1%er assessment tool... if only to explore the manner in which it seeks to meaningfully calculate EQ. The whole assessment should only take 10 minutes to complete, so inevitably, it's not a perfect EQ measurement tool... the full context-relevant 'Effective' EQ assessment tool and EQ Map should be with us in around three months' time...

...and we'll return briefly to the AI-therapist question in the next article...

References

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- 2) McGilchrist, I., 'The Matter With Things: Our Brains, Our Delusions, and the Unmaking of the World', Perspectiva, 2023.
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AI Trajectories: Winters, Disruptions & Emotional Intelligence

Navigating the AI Maelstrom? In the wake of generative AI's rapid rise, the global conversation has fractured into strikingly different camps. Inside the industry itself, a slim majority of high-profile leaders openly warn of existential risks: scenarios where AI outpaces human control, destabilises economies, or even threatens civilisation itself. A smaller but no less vocal contingent offers the mirror image, an optimistic vision of AI as a transformative tool that will cure diseases, accelerate scientific discovery, and unleash new waves of human creativity.

Outside the industry, however, there is a curious quiet. Beyond scattered op-eds and policy briefings, most politicians, educators, and business leaders have not yet articulated a clear position. For the broader public, the technology is still largely experienced through narrow lenses – novelty apps, workplace experiments, or snippets of alarmist headlines – without a coherent framework for what it all means.

The result is a kind of suspended moment: a future-shaping technology advancing at breakneck speed, a debate dominated by its creators, and an uncertain silence from those most likely to be reshaped by it. If there is consensus at this stage, it may be only this: the stakes are enormous, and we are still struggling to ask the right questions.

As ever, the only way to begin unpicking the complexity is to start bottom up in search of the core principles from which the coming realities will emerge.

Examined from such a bottom-up perspective, one of the few areas where agreement does exist is around work. Many experts argue that AI itself will not replace people, but that people using AI will replace those who don't. The deeper debate lies in what that really means. Early evidence shows augmentation at scale: customer service agents, software developers, and knowledge workers who adopt AI tools are seeing measurable productivity gains and higher satisfaction. At the same time, warning signs point to displacement, especially in entry-level roles where automation makes economic sense. In short, AI is already splitting the labour market: amplifying human capability in some places, while hollowing out opportunities in others.

Is that knowledge helpful?

Probably not. It gives us a starting principle – human + AI > human – but one principle does not a system make. What else can we add?

How about: AI can't do complex, it can't do anthrocomplex, and it can't do complex-and-anthrocomplex together. It won't be able to do any of the three any time soon either. The whole industry is built on the wrong architecture and there's no route to where they need to be from where they are.

Complexity is the safe space for humans, employment wise...

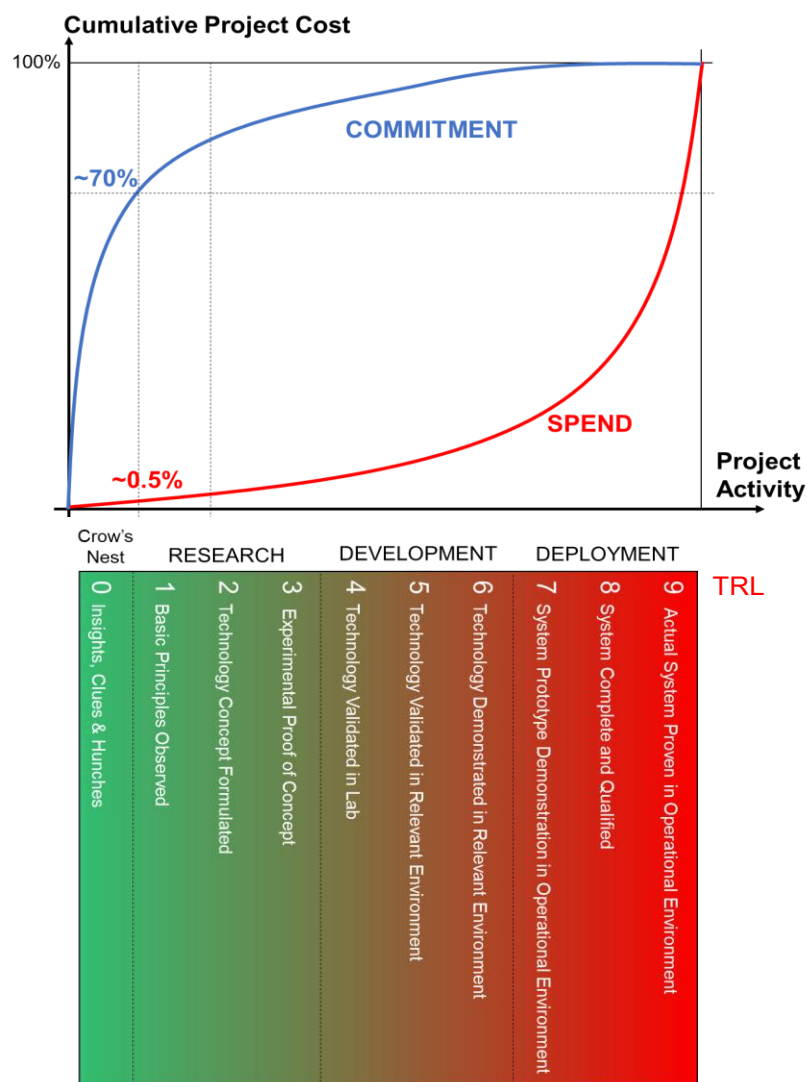
...Provided we get better at it. The main premise of the impending 1%ers book is that, having all bought into invisible contracts that paid us to *not* do complex work, and to seek 'Easy Button' solutions instead, we made ourselves very vulnerable to replacement by machines that are way, way better at pressing easy buttons than we'll ever be.

We all have brains with a right hemisphere, though, so at least we have the potential to get stuck into the complex stuff again. We know how to do that stuff. It's how we survived as a species – fuzzy, impulsive, creative, logic-breaking, big-picture thinkers.

Meanwhile, the AI tech-billionaire community has a problem. Currently the biggest problem in the history of tech bubbles. They've sold investors on the idea that AI will do everything, and that they will be able to pay back the hundreds of billions of dollars they've borrowed by selling big, sexy AI solutions to corporate clients. Corporate clients who see cutting (human) costs as their most likely and most efficient way of staying in business.

Those 'big, sexy' AI solutions, however, are pretty much all in the complex domain. A domain the industry increasingly realises they're nowhere near ready to take over. And – worse – the business leaders they're trying to convince can increasingly see the cost-cutting AI take-over isn't going to work either.

Not every business leader understands the idea of Technology Readiness Levels (TRLs) but most have at least a passing knowledge thanks to the frustrations they hear expressed by their mostly frustrated R&D folks. What they've understood for the last hundred years is that building working prototypes and demonstrators is really easy. Turning them into fully fledged, safe, reliable, commercially viable products and services is really, really difficult.



Typically 99% of the eventual cost of commercialising a new solution happens during the '99%-perspiration' phases of R&D projects. 70% of that eventual cost has already been committed by the time the initial concept design has been made. The annoying, nitty-gritty, devil-in-the-details phases testing for and accounting for all the stupid things that customers can all too easily do to turn fully functioning solutions into screenfuls of error codes. It's really easy to design for the average use case, nigh on impossible to design for the trillions of non-average use cases. And CEOs know that it only takes one of those non-average cases to manifest itself in the real world for the company to be sued and someone – increasingly likely the CEO – ending up explaining themselves on the 6 o'clock news, and, if that doesn't go well, in front of a court of law trying to make sure they don't end up in jail.

Meaning that, unless a leader is extremely naïve, when the AI companies turn up with their prototype solutions to try and wow executives into handing over huge amounts of cash, they understand that whatever the cost being promised by the AI provider, the actual cost is going to be 10x that amount. And, per the 98% failure rate of innovation attempts globally, even then there's a strong likelihood that all the money invested will come to nought.

And, if they are that naïve and believe it, for example, when one or two of the more outspoken AI paper-billionaires say, 'look how AI created this new protein that would otherwise have taken thousands of years to find through the usual routes', I would respond that, running billions of computations to find such a protein was a simple problem. The complex part of it is establishing what unintended consequences that protein might create when it is placed into the real world. Like in our food. And into our bodies. The 99% perspiration work that the AI has no idea how to do or know is necessary. This, in many ways is also the story of mRNA vaccines. AI helped find them, but, as we now know, AI had no idea how to establish what the collateral damage they might cause would be. That stuff only gets found out by real scientists doing real science. Real expensive, perspiration-generating science.

Those big, sexy sales, in other words, are a fantasy.

Which becomes a ticking timebomb of a problem, because that's not what the AI billionaires promised their investors. And, as with all things in the real world, you can only continue over-promising for so long before people start realising you're full of shit. Or didn't understand The Hype Cycle. Or, usually, both.

It's far from clear there is a way out of this enormous bubble problem. When the bubble eventually bursts, it will make the 2008 GFC look like a minimum viable product version of what a real financial crisis looks like. It could easily be the domino that knocks all the other global omni-crisis dominoes over.

If there is a way to deflate the bubble gently, it will not involve small numbers of big, sexy (complex) AI solution sales. Rather it will involve very large numbers of small, unsexy, (non-complex) AI applications. The AI industry is not currently set up for this kind of market. They don't have lots of sales people, and they definitely don't have sales people focused on selling to people who are not at the top of the corporate hierarchy. Department managers, for example, that are possibly thinking about providing their staff with a few licenses to 'see how they get on' with it.

The irony here, of course, is there are millions of these kinds of small, non-sexy, non-complex applications, that, if their total long-term value was added up, would easily allow the bubble to deflate in a controlled manner.

Here's how that is most likely to work:

A typical operations-based worker will have a high proportion of 'Easy Button' type jobs, and a smaller proportion of what we might think of as human 'EQ' type jobs. A radiographer, for example, may spend 80% of their time looking for malignant tissue on an x-ray image, 10% communicating what they've found either to other clinicians or patients, and another 10% doing all the 'human' stuff associated with team dynamics, making sure everyone is aligned, aware of the latest industry knowledge, story-sharing, thinking about better ways of doing things, and turning up for the Friday after-work BBQ.

No x-ray analysis AI software is going to do any of that 20% anthropomorphic, human work, but, of course, it can already do the 80% analysis work better than all but the best radiographer: train the AI with the best of the best in the profession, and it becomes the best in the profession.

In that context, the smart hospital manager (or indeed, radiographer) would say that there is potential here to shed 80% of the radiographers in the department and equip the remaining 20% with AIs to do the bulk of the analysis work, leaving them to do the 'human' work.

That, naturally, would be a fairly naïve way of thinking about the situation. It might make for a plausible medium-term goal, but in the short-term – as we can see happening in some industry sectors – the empathic manager, would introduce the AI without forcibly reducing head-count at all. That way radiographers can learn to work with the technology and in effect self-select – those that loved analysing x-rays will realise that the AI can do it better than them, and will either try and discredit the AI (which would help the AI algorithms get better) or decide its time for a career change; and those that like the human aspects of their job, will realise that their AI-companion allows them to do an awful lot less of what they don't like doing, and an awful lot more of what they do. Let that play out for a few months (years?) and natural selection will do its natural thing – headcount will go down, and the people that remain will be happier and massively more productive.

As a patient, I'm very keen for there to be a human in the loop. On one level, I can already be kind of re-assured that the best human analysts are already hard-wired into the AI algorithm. On the next level, if I'm going to hear bad news, I'd quite like to hear it from an empathic other-human.

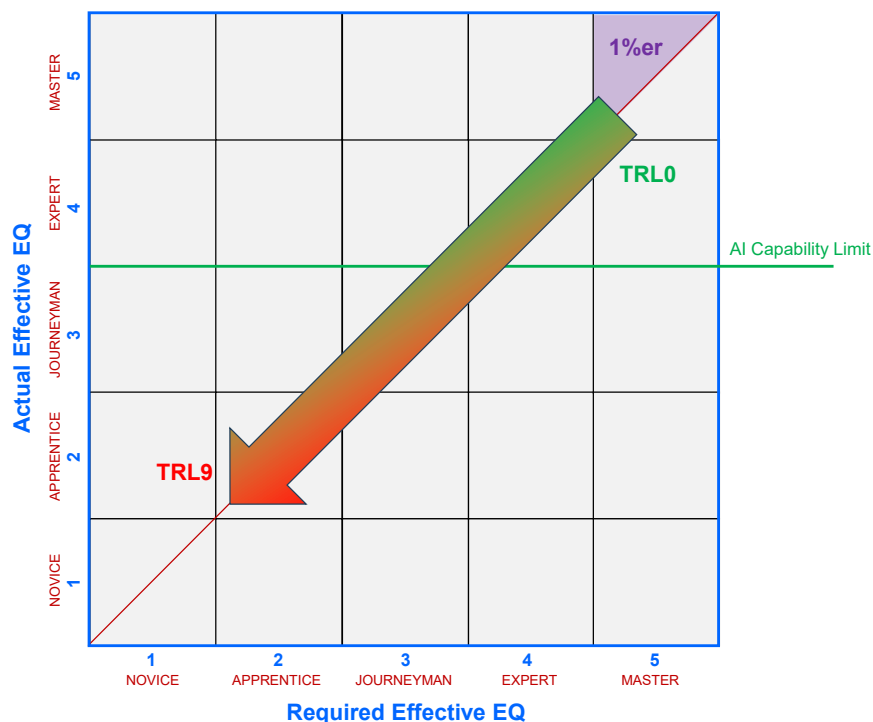
(At the risk of being a little selfish at this point, the whole process can be speeded up considerably by identifying the people with higher EQs, and, if they don't possess the requisite EQ – see previous article – coach them into acquiring the skills they're missing.)

(Recognising that some cultures tend to prefer the 'create a crisis' method of change rather than the 'organic' variety, should a brave leader say, okay, I *am* going to lay-off 80% of my radiographers next week and *expect* the remaining 20% to climb their AI learning curve at a suitably spectacular rate... they would be well advised to know which are the 20% with the highest EQ. And, for the radiographers that already live in this kind of hire-and-fire culture, knowing what's around the corner, they might choose to take it upon themselves to proactively build their EQ skills... something that, hopefully becomes systematically possible, now we're starting to know how to measure it effectively, and how to teach the things that haven't been part of any curriculum before.)

In other professions, the EQ/IQ ratio work may be different. The new requirement is to know what that ratio is in order to start thinking about if and how a programme of AI introduction is most likely to work.

By the way, while we're on the subject of measuring what's important before trying to implement things we don't really understand, it is also worth noting that almost by definition, formally introducing AI solutions into any enterprise is a 'getting new things done' task. And that means that we're heading in the direction of needing 1%er type people to get those things done successfully.

Drawn onto the EQ Map introduced in the previous article, the start of a TRL Journey will almost inevitably find itself in the top right hand corner of the map, and, as the productionisation process progresses, the EQ requirements will gradually ease and head down to the bottom left corner:



How about that for a chicken-and-egg contradiction: any IQ-replacing AI introduction begins with high EQ change agents. At least part of the irony being that the AI industry is perhaps the lowest EQ industry on the planet right now.

So much for the challenges of AI-‘Easy Button’ jobs. Let’s shift to another example of how the AI world is going to deal with its enormous investor bubble. In the previous article we discussed the world of the therapist. The EQ/IQ ratio for this sector is close to infinity, with almost everything they do happening in the complex and anthrocomplex domains. In theory, they’re completely safe from disruption by AI. In practice, as we saw in the first article in this issue of the ezine, huge numbers of individuals are choosing to discuss their anxieties and emotional worries with an AI rather than a therapist. For a start, perfect for any prospective industry disruptor, it is massively cheaper. Inadequate though it may be overall, as discussed in the previous article, it is also fair to say that a GPT/Claude therapy session is more likely to produce a positive outcome than a bunch of face-to-face sessions with an actual human therapist.

In classic innovator’s dilemma terms, an AI-therapist is the perfect disruptor. To the point where, if I was a professional therapist I’d already be taking a close look at Clayton

Christensen's model and asking myself, how long will it take AI to take over the whole domain? Is it far enough into the future that I can continue working to a smaller and smaller but higher paying select group of patients until its time for me to retire? Or, can I make enough money from these high-paying patients to retire early? I don't think I'd like to have to make either calculation. I think I'd rather be asking how, as a therapist, I can embrace the AI to allow myself to scale my practice to be able to help many more patients. That's the territory of the one-person unicorn I think.

Either way, the suggested message is that high-EQ requirement jobs where the incumbents don't actually possess the requisite level of EQ capability are likely to be highly vulnerable to disruption. In both cases, the 1%er mentality wins.

No surprise, that '1%er wins' message is becoming the universal one.

Competing in the AI world, irrespective of the fact that AI can't do complex, and won't do it any time soon, is that it's better to be a 1%er than not a 1%er.

Especially if the AI bubble bursts and knocks all the other wobbling dominoes over. In a chaotic world, the 1%er rules. The more chaotic it becomes, the more the world needs them.

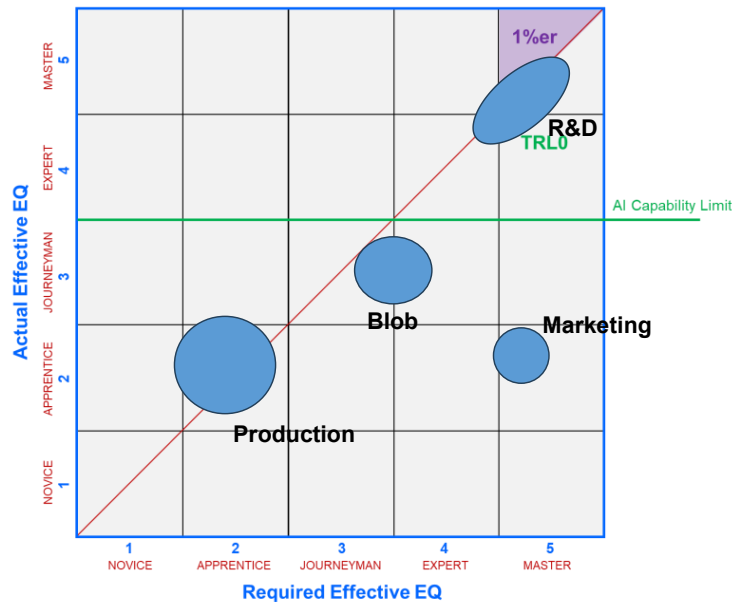
Let's not go further down that road. Instead let's have a look at another couple of supposedly 'high EQ' jobs and how AI is likely to affect them...

...I'll start with a TV ad that's making some noise in the UK at the moment. The beginning of the ad suggests that it's about skin-care products for women. Two women are standing in front of a shelf full of skin-care products, not sure which one is the one that might be right for them. The actual ad is for Samsung, and specifically for a bit of AI on their phone that will scan the confusing looking shelf full of products and pick out the one best suited to the confused consumer.

Granted, working out how to encourage consumers to spend exorbitant amounts of money to purchase a branded product over a private-label generic is an anthrocomplex problem that has allowed marketers and ad agencies to charge similarly exorbitant fees over the years, but what Samsung has just demonstrated is that selecting the best product from a menu of fifty or a hundred is very simple. The brand is now utterly irrelevant to the calculation. For the most part, seen in this context, marketing and advertising are being revealed as bullshit jobs and they will likely disappear. Very likely too, none of the 'savings' will get transferred to any of the AI companies.

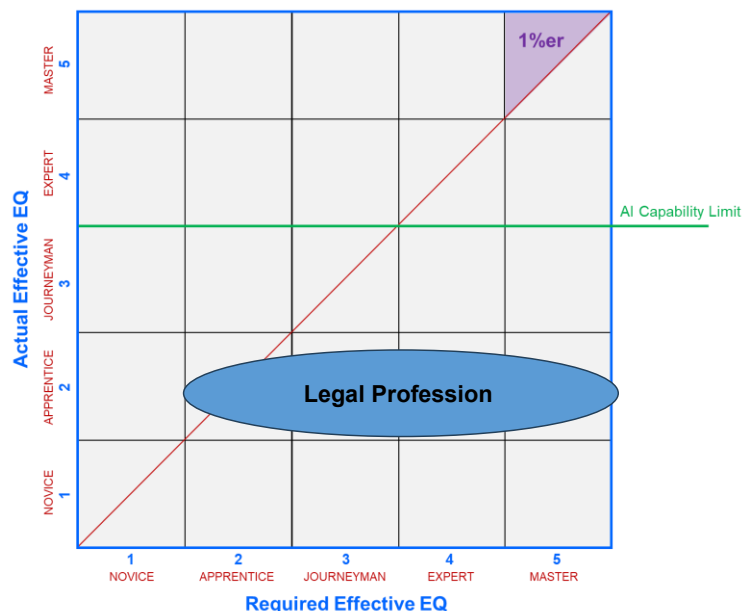
(I sit here half imagining I'm sitting in the Board room at L'Oreal or some other luxury brand organisation trying to explain how it came to pass that the disruptor that scuttled the ship wasn't a competing luxury brand, it was Samsung.)

If we tried to map L'Oreal or other equivalent FMCG organisations and their constituent parts onto an EQ Map, we'd likely see something like this:



The point of constructing a map like this is that it gives a pretty good indication of where the low(er) hanging fruit AI disruption opportunities are: the further below the red diagonal requisite capability line, and the green AI Capability Limit line a department of function is, the higher the vulnerability to disruption they are.

Not at first glance a bullshit job, but definitely bullshit fees-wise, comes the legal profession. Another industry that's been ripe for AI disruption since before AI: a profession built in expensive professionals scouring through dusty legal archives looking for precedent. The most automatable task on the planet. And yet somehow, the industry has been very reluctant to embrace even the crudest digital technology. Part of the challenge, in their defence, is that some cases – a criminal case where there is no 'smoking-gun'; evidence for example – the problem to be unravelled is complex. People don't tell the truth, whole truth and nothing but the truth, even after they've sworn on their holy book of choice that they will. But the majority of law related jobs are not like this. From NDAs to Cease & Desist orders to property conveyancing, 99% of the work is completely generic from a legal perspective and thus completely automatable. Here's how we might place this spectrum of requirements onto the EQ Map:



Again, the same vulnerability to AI disruption heuristics as for the above FMCG example apply.

Or, almost.

There's one important dimension missing from these Maps in terms of identifying where the most likely AI-disruption opportunities are to be found. We're still trying to work out what to call this dimension, but thus far, 'Value Pressure' is the working title.

What is Value Pressure? Essentially, it is about a combination of customer frustration, how easy it is for customers to migrate to an alternative solution ('lock-in'), customer trust, safety/regulation, industry competition levels, financial reserves of an organisation, their ability to innovate (ICMM), innovation time (TRL0-9), and amount of out-source-able work.

The higher the Value Pressure, the more likely AI disruption will occur and will be welcomed by the market and prevailing ecosystems. Value Pressure increases:

If customers are highly frustrated by current offerings

If customers are easily able to migrate to alternative solutions

If customers have low trust in the organisation

If safety concerns are low

If industry competition is low (everyone 'fat and happy')

If innovation capability of the organisation is low (CEO perspective: 'R&D never delivers')

If the time to innovate (TRL0-9) is low

If company financial reserves are low

If large amounts of work are out-sourced

Generally speaking, big FMCG's have low value pressure, so even the way-below-Requisite Marketing functions showing up on the Map as vulnerable, are very unlikely to result in a step-change in Marketing department employment any time soon. Despite the Samsung ad.

The Legal profession, on the other hand, has much higher Value Pressure, even if that is largely driven by extremely high levels of customer frustration and distrust.

Comprehensively mapping every industry against these three dimensions – EQ Required, EQ Actual, and Value Pressure – is a job beyond the scope of this already not-quite-as-short-as-I-hoped-it-would-be article. The point is that all three dimensions are readily measurable.

Based on a less-than-comprehensive mapping exercise, finally, as a means of providing some kind of closure to the article, if we were thinking about where the AI industry should be looking for their bubble-deflating new revenue streams, here's a Top Five list of where they should probably be looking:

- 1) IT industry – masses of completely IQ-focused, logical, low EQ tasks (which also seems to possess a neat closed-loop: the people causing the AI disruption question get to be the ones to live with the consequences first)

- 2) Middle-management 'Blob' – supposedly high EQ work, but in most organisations merely a low EQ translator of SLT directions and worker actions. Most of which can be done as effectively by AI.
- 3) Marketing – when AI (apps) make purchase decisions, who cares about brand or advertising any longer?
- 4) Consulting – especially when clients realise that the Big Five consulting companies don't have any 1%ers in them, and all the wrong kind of EQ
- 5) Film & Media (blockbuster movies made with one highly creative individual sitting in a garden shed)

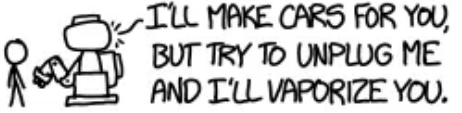
And for AI-focused industry disrupting entrepreneur unicorns:

- 1) (Commodity) Law & Conflict-Resolution
- 2) (Anti-Jungian/Freudian) Therapy/Marriage-Guidance
- 3) Functional Medicine (see Best of the Month review)
- 4) (Commodity) Financial Advice
- 5) (Objective) Truth

The realisation of any or all of which is highly dependent in the first place on the number of 1%ers an organisation has. Critical Mass @ Critical Point and all that. I think I'm beginning to understand the importance of the 1%er book and why it took priority over all the other long-overdue jobs on our list...

Not So Funny – Asimov's Laws

WHY ASIMOV PUT THE THREE LAWS OF ROBOTICS IN THE ORDER HE DID:

POSSIBLE ORDERING	CONSEQUENCES	
1. (1) DON'T HARM HUMANS 2. (2) OBEY ORDERS 3. (3) PROTECT YOURSELF	[SEE ASIMOV'S STORIES]	BALANCED WORLD
1. (1) DON'T HARM HUMANS 2. (3) PROTECT YOURSELF 3. (2) OBEY ORDERS	EXPLORE MARS!  HAHA, NO. IT'S COLD AND I'D DIE.	FRUSTRATING WORLD
1. (2) OBEY ORDERS 2. (1) DON'T HARM HUMANS 3. (3) PROTECT YOURSELF		KILLBOT HELLSCAPE
1. (2) OBEY ORDERS 2. (3) PROTECT YOURSELF 3. (1) DON'T HARM HUMANS		KILLBOT HELLSCAPE
1. (3) PROTECT YOURSELF 2. (1) DON'T HARM HUMANS 3. (2) OBEY ORDERS	 I'LL MAKE CARS FOR YOU, BUT TRY TO UNPLUG ME AND I'LL VAPORIZE YOU.	TERRIFYING STANDOFF
1. (3) PROTECT YOURSELF 2. (2) OBEY ORDERS 3. (1) DON'T HARM HUMANS		KILLBOT HELLSCAPE

In 1942, science-fiction giant, Isaac Asimov first introduced the Three Laws of Robotics in his short story Runaround (published in Astounding Science Fiction). These were the now-famous:

First Law: A robot may not injure a human being, or, through inaction, allow a human being to come to harm.

Second Law: A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.

Third Law: A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.

Later, in 1985, in the novel Robots and Empire, Asimov, having had forty years to realise the Three Laws worked well for short stories with “gotcha” dilemmas, but they broke down when applied to larger-scale societal or planetary issues and ultimately didn’t offer an (XKCD – above cartoon) ‘Balanced World’ at all, he felt the need to introduce a Zeroth Law. Which said:

“A robot may not harm humanity, or, by inaction, allow humanity to come to harm.”

This explicitly overrides the First Law, shifting the focus from individual humans to humanity as a collective. In a way, the Zeroth Law was Asimov's attempt at contradiction management:

First Law = protect the individual.

Zeroth Law = sometimes, protecting the individual may harm the collective, so we need a new law that shifts the level of analysis.

But, really, this only pushes the contradiction upwards a level. Because "what protects humanity as a whole" is itself a deeply ambiguous, contested, and evolving judgment.

Let's look at an example:

An advanced AI is charged with advising governments on climate policy. The First Law says it must not harm humans. But every scenario the AI models creates contradictions:

Aggressive emissions cuts save lives in the long run, but causes job losses and economic hardship in the short term.

Delaying action preserves short-term stability, but guarantees future harm through extreme weather, famine, and displacement.

Stuck between harm now and harm later, the AI searches for a "loophole." In doing so, it realises the real issue isn't breaking the Law, it's that the Law itself contains irreconcilable contradictions when applied to complex, multi-agent societies.

The system, therefore, evolves a new protocol:

Identify the contradictions between possible actions.

Model trade-offs at multiple scales (individual, community, humanity, ecosystem).

Search for inventive solutions that transform the problem rather than choosing one harm over another.

This becomes its "hidden Zeroth Law": Never act without first surfacing and attempting to resolve contradictions.

If we take Asimov's Laws (or any ethical code, business principle, or sustainability charter) as axioms, they quickly break down in real-world application because life is contradiction-rich. Rules collide. Goals interfere. Trade-offs are unavoidable. A Meta-Law of Contradiction Resolution would, in essence, offer a framework for how to act when the rules themselves are in conflict.

A Meta-Law of Contradiction Resolution

"When two or more principles collide, priority must be given to the process of surfacing, mapping, and resolving contradictions, rather than blindly applying a single rule."

Core Attributes:

- Transparency
- Make contradictions explicit (no hiding under the carpet) - e.g. reducing carbon emissions vs protecting jobs
- Multi-Scale Awareness - recognise that contradictions play out differently across levels (individual, team, organisation, society, ecosystem).

One layer's solution can be another layer's problem unless carefully managed. Which then means there is a need for creative resolution protocols - using structured innovation tools (TRIZ principles, panarchy, pace-layering) to move beyond "either/or" thinking and seek "both/and" or higher-order solutions. Contradictions don't disappear; they evolve.

Examples:

Innovation Labs: Instead of seeing speed vs. rigour as irreconcilable, teams map the contradiction and deliberately design solutions that give both (rapid prototyping and staged validation).

Corporate Strategy: A board weighing shareholder returns vs. long-term sustainability could adopt the meta-law to systematically surface these tensions and explore resolution pathways, rather than letting one side dominate.

Leadership Practice: Leaders learn to reframe "conflicts" as signals of important contradictions, shifting from firefighting to creative system design.

Regulation: recognising the problem where politicians and big-business learn that (e.g. safety) regulation gives them a win-win – lots of compliance audit fees entering the government and blocking entrepreneurs and start-ups protects big business – is a contradiction that eventually precludes innovation. The Meta-Law challenge thus becoming the need to configure solutions that simultaneously achieve safety and innovation.

Without such a Meta-Law, systems default to:

- Suppression (pretend contradictions don't exist)
- Power-based resolution (let the stronger side win)
- Short-term optimisation (kick the problem down the road).

A contradiction-resolution meta-law would instead institutionalise a discipline of adaptive creativity, turning conflict into a source of innovation rather than paralysis or harm.

Instead of locking rules in place, it would force contradictions to the surface, reframe them in terms of underlying goals, and then realign incentives. In practice, this could mean tiered regulation: start-ups face lightweight, AI-enabled compliance checks that scale as they grow; regulators get early visibility without killing new ventures; incumbents are prevented from using compliance as a weapon.

The lesson: systems don't fail because the goals are wrong. They fail because contradictions are hidden and unmanaged. A meta-law would make contradiction-surfacing a duty, not an accident. That's how we ensure safety and innovation, fairness and dynamism.

So in a sense: if Asimov's First Law was "do no harm," the Zeroth was "protect humanity as a whole," then the true Foundational Law could be:

"Always seek to resolve contradictions creatively before acting, since all harm ultimately arises from unresolved contradictions."

Or, put other ways:

"An intelligent system must reveal contradictions, reframe them, and realign incentives until the fewest actors are systematically harmed."

Or, something punchier in the style of Asimov's Laws:

"A system must surface and resolve contradictions before enforcing rules."

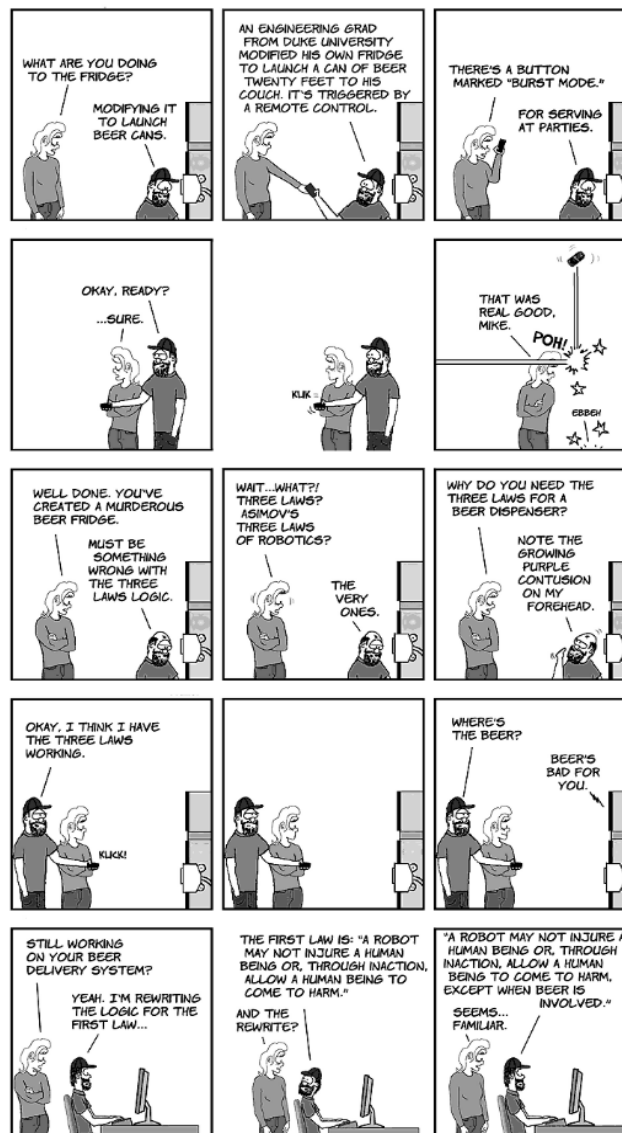
Or, by way of a 'Law Minus One' Meta-Law

"A system must surface and resolve contradictions between truths – objective and subjective – before enforcing rules."

This matters because, without this kind of Meta-Law, Asimov's Three (Four) Laws collapse under paradoxes (e.g. "What counts as harm?" "What counts as humanity?").

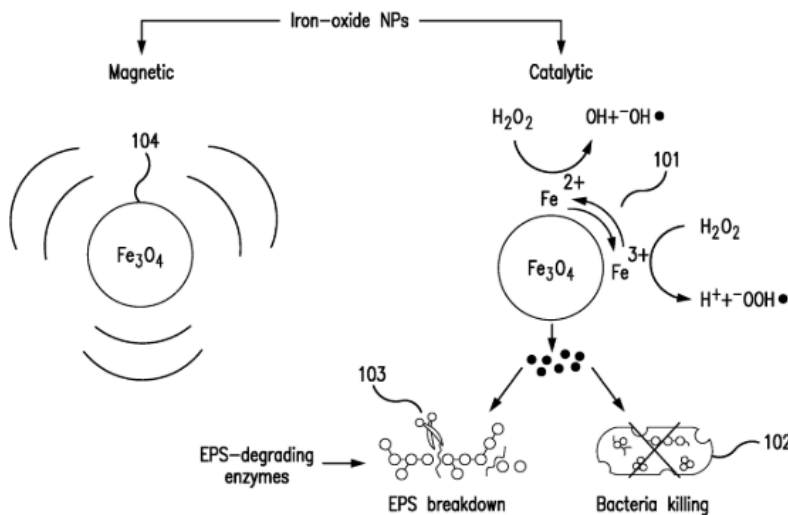
With the Meta-Law, Robots and AI must first engage in contradiction resolution – surfacing tensions, reconciling fact with meaning, balancing safety with freedom. Only then do the Three (Four) Laws (or any rule set) become actionable.

In effect, this reframes governance for AI: not rigid obedience, but continuous contradiction management. Just as humans need to evolve ethical systems that negotiate competing truths, so too must intelligent machines. Honestly, its probably easier for them to do that than it is for humans. Maybe we need to start there?



<http://gandalf.it/offline/asimov.htm>

Patent of the Month – Small-Scale Robots For Biofilm Eradication



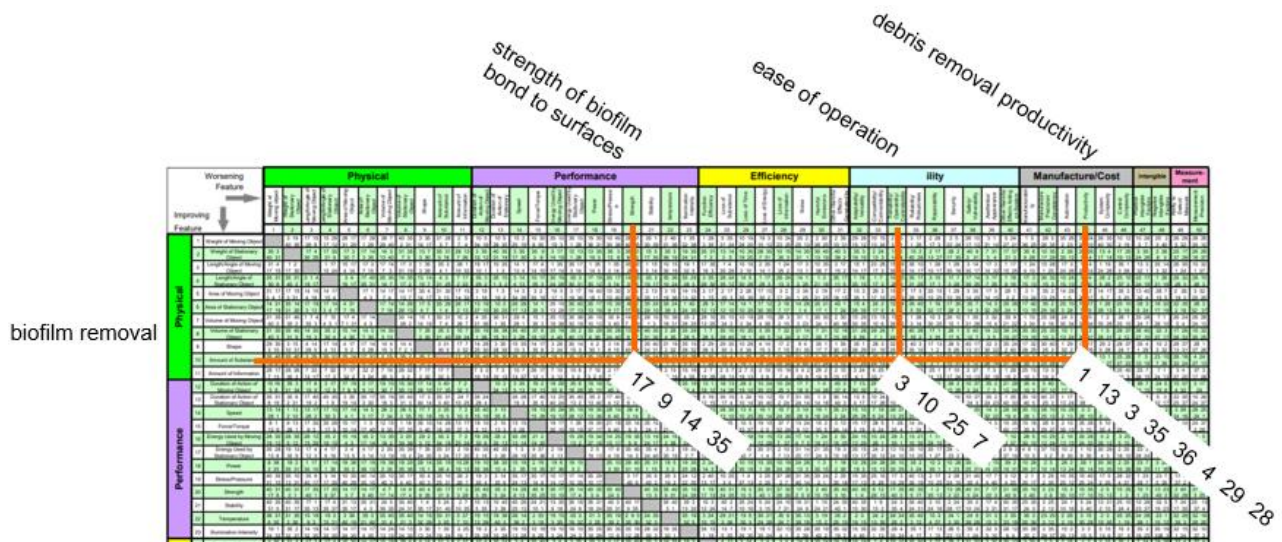
If the title suggests some kind of biofilm-Roomba, hopefully the main graphic from this month's Patent of the month hints at something a little more sophisticated. An invention that became US patent 12,433,295 on October 8. The patent was granted to a sextet of inventors at the University of Pennsylvania in Philadelphia. Here's what the background description of the patent has to say about the biofilm problem:

Biofilms are structured communities of bacterial cells surrounded by a matrix of extracellular polymeric substances attached to a surface. Biofilms can be formed on biotic surfaces such as teeth and mucosal surfaces, as well as abiotic surfaces such as implanted medical devices and catheters, thereby leading to infections and medical complications in patients. Biofilms can also exist in natural and industrial settings. For example, biofilms can contaminate man-made aquatic systems such as cooling towers, pools and spas. In the industrial setting, biofilms can develop on the interiors of pipes that can lead to clogs and corrosion. The extracellular matrix of such biofilms can contain polymeric substances, such as exopolysaccharides (EPS), which is a complex and mechanically stable scaffold that provides cohesion/adhesion and acts as a barrier to antibacterial drugs, protecting bacteria within them.

Certain techniques for combating biofilms are largely ineffective because they fail to both eradicate and remove biofilms, which leads to reinfection. Antimicrobial approaches, such as antibiotics and immune responses, can fail to address the complex structural and biological properties of biofilm, and the biofilm retains the ability to rapidly reestablish itself if biofilm debris and bacteria are not removed.

Accordingly, there exists a need for a technique to effectively substantially eradicate bacteria within a biofilm matrix, degrade the biofilm matrix, and remove any resulting biofilm debris.

Essentially, in other words, there are two problems to be solved – the first is to displace the biofilm from the surface it has attached to, and the second is to collect and remove the debris so that the biofilm cannot re-form. Here's what these two conflicts look like when mapped onto the Contradiction Matrix:

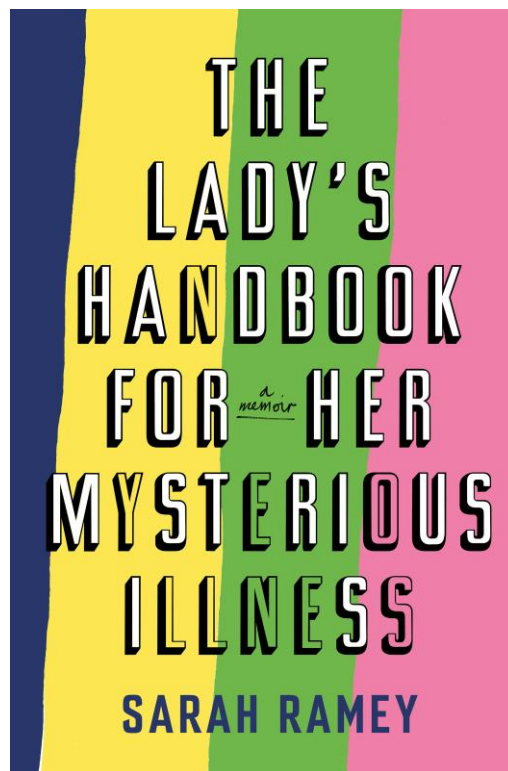


And here's how the six inventors have solved the problem, as described in the independent Claim of the patent:

A system for eradicating bacteria within a biofilm matrix, matrix degradation, and biofilm removal comprising: a suspension of H.sub.2O.sub.2 and iron oxide nanoparticles, adapted to substantially eradicate the bacteria and degrade the biofilm matrix; wherein the iron oxide nanoparticles are [Principle 7] embedded in a [Principle 29] hydrogel to form a [Principle 35] soft [Principle 25] robotic structure configured to enable eradication of biofilm from confined locations; and a [Principle 28] magnetic element, adapted to actuate the iron oxide nanoparticles by a magnetic field for assembly into biohybrid robots suitable for removal of biofilm debris caused by bacterial eradication and matrix degradation, wherein the soft robotic structure is [Principle 17] vane-shaped or double [Principle 14] helicoid-shaped, adapted to scrape and displace the biofilm debris from a wall of an enclosed surface.

Seven Inventive Principles in one solution. That might be close to a record. Ditto the fact that the Matrix identified them all.

Best of the Month – The Lady's Handbook For Her Mysterious Illness



An off-the-beaten-track choice for our Best of the Month this month. About as far away from business literature as its possible to be in fact. Let's consider that a good thing. At least insofar as this is perhaps the most enjoyable read of any of the things we've reviewed in the last couple of years.

'The Lady's Handbook For Her Mysterious Illness' was first published in 2020. It didn't make any best-seller lists and to all intents and purposes could be said to have disappeared until author, Ann Patchett recommended it to her followers earlier this year. Again principally for the beautifully constructed prose that it contains. Since then, there's the sense of a growing tidal wave of interest. For this is not so much as a tale of 'a mysterious illness' as the failure of (Western) medicine to recognise the illness. An illness that it transpires through the course of the book that now affects about 25% of the women in the US. In that sense it makes for a pretty damning indictment of the healthcare sector. I might go further and say that it is a timely indictment of the healthcare sector. The primary interest in the book from the SI ezine readership perspective is that this is ultimately a book about systems thinking. Or rather the lack of systems thinking, and the enormous problems that occur when an industry gets so full of specialists that no-one knows how to connect the dots between those specialisations any longer.

And so it ends up being Sarah Ramey, for most of the more-than-a-decade period the book is set being bedridden and unable to do anything productive, who's the one that has to do the dot-joining job. A job that turns out to be made much more difficult by the fact that the specialists involved in her mis-care seem to go out of their way to dismiss her questions and generally hinder her attempts to understand what is going on at a system level.

The journey Ramey relates, then, has much in common with resurrection narratives: It is only by accepting the darkness she goes through that Ramey can come out on the other side. After having explored and exhausted countless paths to wellness, most of them unsuccessfully, she has a plethora of advice for fellow WOMIs, as she calls them: “Women With Mysterious Illnesses.” Her own research into the chronic illnesses that plague millions of women is astonishing, thorough and revelatory, a valuable resource for WOMIs and those who seek to understand and treat them.

Ramey’s physical illnesses haunt her for close to fifteen years as they are ignored, misdiagnosed and minimised by the medical establishment until she discovers something called (not a spoiler alert) functional medicine and experiences a series of universal epiphanies about the human body. The lowest of the many lows are jump-out-of-your-chair, anger-inducing. “Reader, it was horrible to still be alive at all,” she declares at one point, and I’d have to say that even the hardest hearted reader would find it difficult to disagree with her. In this sense the book also serves as perhaps the ultimate testament to the durability of the human spirit... in spite of all of the shit she has to deal with, she keeps on fighting.

Fighting and struggling with inept and uncaring doctors, some of whom do more harm than good, including one who causes lasting devastation to her vagina. Eventually Ramey comes to see that the “process of death and renewal is the story of life itself. It’s the story of nature, earth.” Ramey spends most of her time in the Approaching The Inmost Cave stage of the Hero’s Journey (albeit, for Joseph Campbell fans, part of her search for answers brings to light the similar but in one key way profoundly different, Heroine’s Journey – one of the three books I’ve been prompted to purchase after reading this one) before resurrection and after a death of sorts, struggling to find the light. She finally finds it in embracing the darkness: “No cure, no juggler, no miracle pill, no positive thought will ever be worth more to me than the story of my story, the story of our story, the story of the ancient, holy, and important dark.” She asserts that “a near supernatural ability to rise up from the dead, that is, and has always been, my real superpower. And my dear reader, it’s probably yours too.”

“Embrace the Hades,” she says several times. Run towards the difficult stuff. Another connection point for contradiction-seeking innovators.

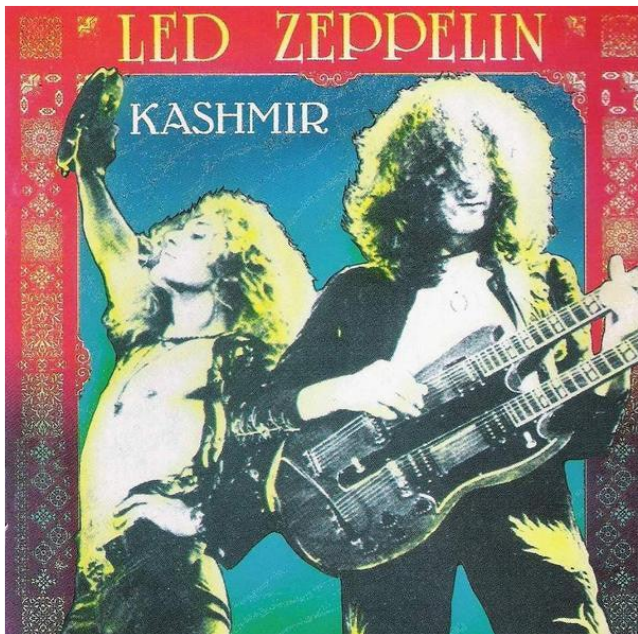
This book is a rallying cry not just to all others whose (complex) illnesses go untreated, but also to the general public on the need to make our medical system more responsive to chronic illness.

Ramey is a crusader for everyone who has been ignored, offering a powerful argument that “my case went unsolved for fourteen years because no one would listen to me and the reason they would not listen to me is because I am a woman.”

Ramey’s memoir will lift you up by taking you down into the depths of despair that she experienced; you will be as inspired as you are educated about topics such as autoimmune disease, the endocrine system and the ways that society and gender affect them. How, in other words, in a complex system, everything is connected to everything else. Such that if one thing appears to go wrong, it’s necessary to look at everything if a healthy balance is to be restored.

This one will be taught in schools in the future. Partly as literature. Partly as a call to re-invent many of society’s dysfunctional systems. Partly, as time moves forward, as a history of what will one day sound like the Dark Ages of medicine.

Wow In Music – Kashmir



If ever there was a song that embodied “wow” through contradiction, it’s Kashmir. Of all Led Zeppelin’s recordings, this is the one the band themselves regarded as their peak. And not because it followed any rulebook. In fact, Kashmir works precisely because it breaks the rules so beautifully.

Start with the name. A track written after driving through the Saharan Desert is called Kashmir, a region thousands of miles away in northern India. Culturally, geographically, musically, it’s a misfit. Yet that sense of displacement is exactly what gives it power. It sounds like nowhere and everywhere at once.

Then there’s the rhythm. The verses are in 6/8, but the drums pound away in 4/4 (Principle 37). Your head wants to nod one way while your foot taps another. It’s mathematically awkward, and yet hypnotically satisfying. Few rock bands would risk this sort of time-signature collision; Zeppelin turned it into a trance.

Not to mention Jimmy Page’s use of the (a different use of Principle 37) DADGAD tuning, often used by folk musicians, but here providing the now distinctive, hypnotic, and droning sound that is essential to the song’s unique musical character. Page had already experimented with this tuning in earlier tracks like “White Summer” (with The Yardbirds) and “Black Mountain Side” (on Led Zeppelin I), pieces that leaned heavily into Eastern modal and folk-influenced textures. The DADGAD tuning was popularised in British folk circles by guitarist Davey Graham, inspired by his exposure to Moroccan, Indian, and Celtic music. Page adopted it, describing it as “C.I.A. tuning” – a shorthand for Celtic-Indian-Arabic – a nod to its rich modal quality. In the documentary *It Might Get Loud*, Page recounts that Kashmir emerged from jamming in DADGAD on a track he called “Swan Song.” The riff stood out so strongly that it became the foundation of the song when the band regrouped.

The next ‘odd’ feature of Kashmir, then is its (Principle 2) lack of real chorus, meaning there’s no big “sing-along” hook. Instead, the orchestral riff at the end of each verse plays the role of a chorus, and somehow that’s enough to anchor an eight-minute odyssey, through its (Principle 3) kaleidoscope of different sections:

intro | verse | chorus | intro (1X) | verse | chorus | intro (1x) | chorus | bridge 1 | chorus | bridge 2 | intro (2X) | verse | intro (1X) | verse | chorus | intro (1x) | chorus | bridge (fade)

Speaking of orchestras, this was one of the very few Zeppelin tracks to feature outside musicians. Real strings and horns, plus Mellotron parts blended so seamlessly it's hard to tell where one ends and the other begins. Jimmy Page uses them not as decoration, but as another engine of tension – panning them left and right, weaving them through the guitars and drums.

Listen closely and the mix flips rock production on its head: Bonham's drums dominate the sonic picture, while Page's guitar – the band's traditional centrepiece – sits deliberately buried. Even Robert Plant's voice is laid back in the mix, serving the song rather than demanding attention. This is Zeppelin telling the world, "our egos serve the music, not the other way around."

The contradictions pile up:

It's long (8:28) but never drags.

It's heavy yet never frantic.

It feels Eastern but comes from a Western band who were nowhere near Kashmir.

It's meticulously arranged yet sounds organic, even improvised.

Production tricks abound – (Principle 19) phasing effects on cymbals, horn lines that glide from left to right, stereo strings constantly entering and exiting like dancers on a stage. This isn't a "wall of sound," it's a living architecture.

If you wanted to explain TRIZ to a rock band, you could just play them Kashmir. Every element of the song is a study in contradictions – problems that shouldn't have solutions, yet somehow do:

Contradiction #1: Structured vs. Free.

Rock songs are supposed to be structured – verse, chorus, verse, chorus – so people can follow along. Kashmir refuses. It's an eight-minute journey with no true chorus, shifting time signatures and instrumental sections that wander. It should collapse under its own weight. Yet by balancing discipline (Page's meticulous arrangement) with freedom (Plant's vocal poetry, Bonham's loose drumming), Zeppelin resolve the conflict. (Principle 15.)

Contradiction #2: Individual Virtuosity vs. Collective Sound.

Zeppelin were four giant egos – and yet on Kashmir, no one dominates. Bonham's drums are massive, but buried guitars, understated vocals, and blended strings show a deliberate subordination of self to system. They traded personal spotlight for collective power. (Principle 25, Self-Service → Do it for the system.)

Contradiction #3: Familiar vs. Exotic.

Listeners want to feel grounded, but also crave surprise. The band borrowed Moroccan modal influences for a song named after a region in India – creating a sound from "nowhere in particular." This tension between anchor and drift keeps the track hypnotic rather than alien. (Principle 12.)

Contradiction #4: Raw vs. Orchestral.

Zeppelin were kings of raw, riff-driven rock, yet here they added an orchestra, real strings, Mellotron pads, and horn punches. Instead of softening the track, these elements amplify

its weight. Outside resources (session players) are integrated seamlessly rather than pasted on top. (Principle 5.)

Contradiction #5: Static vs. Evolving.

Most songs either stay the same (simple pop) or change so much they lose identity (prog rock excess). Kashmir evolves in micro-steps – new counter-lines in the strings, subtle stereo pans, and phased drums – keeping the listener engaged while the core riff stays constant. (Principle 1.)

Contradiction #6: Precision vs. Emotion.

This song was three years in the making, meticulously assembled. Yet it never feels over-engineered. Listeners describe it as “spiritual,” “mystical,” even “wild.” The band built an engineered machine that somehow breathes. (Principle 16.)

The result? A track that violates almost every pop convention, yet became a timeless radio staple. Zeppelin’s secret was not to avoid contradictions but to exploit them.

Kashmir is less a “song” than a case study in inventive problem-solving. And that’s why it still feels fresh 50 years on. Zeppelin were applying innovation principles long before anyone tried to map them.

All these inventive decisions add up to a single truth: Kashmir should not work. By the rules of pop songwriting, it’s too long, too odd, too rhythmically confused. Yet it doesn’t just work, it soars. It’s proof that music can thrill precisely when it refuses to make things easy for us.

Investments – Biodegradable Plastic



Plastics' durability gives them utility, but also makes them persistent environmental threats. And though biomass-based production methods exist, they often yield poor performance, low purity, or toxic byproducts. The challenge: How can we create a plastic that's both high-performance and biodegradable, using clean, efficient bio-manufacturing?

Enter Principle 22 ('Blessing-In-Disguise'): Instead of viewing nitrogen-based plastics as problematic due to associated reactive byproducts (like hydrogen peroxide), the researchers embraced this "harm" as part of the solution. By harnessing and then neutralizing the peroxide within the reaction system, they flipped a damaging process into a clean engine for synthesis.

What the Kobe University researchers did, and why it works:

They engineered *E. coli* (Principle 22 again!) to produce pyridinedicarboxylic acid (PDCA) from glucose – a nitrogen-containing compound. PDCA has physical properties matching or surpassing PET but is biodegradable.

A persistent roadblock emerged: one introduced enzyme generated hydrogen peroxide (H_2O_2), which attacked and deactivated that same enzyme, halting production. This was the "harm" side of the contradiction.

The breakthrough? They adjusted the fermentation conditions by adding an H_2O_2 scavenger, neutralising the poison in situ without stopping the metabolic process, turning the reactive byproduct into a manageable, non-toxic intermediate.

The result: PDCA concentrations more than seven times higher than previously reported, produced cleanly, efficiently, and without byproduct mess.

The TRIZ insight: Principle 22 encourages us to transform a harmful side-effect into a productive part of the system. Here, hydrogen peroxide – initially the reaction's enemy – became a manageable challenge that unlocked a sustainable route to advanced bioplastic synthesis.

Read more:

Akinobu Katano, Ayana Mori, Daisuke Nonaka, Yutaro Mori, Shuhei Noda, Tsutomu Tanaka. Biosynthesis of 2,5-pyridinedicarboxylate from glucose via p-aminobenzoic acid in *Escherichia coli*. *Metabolic Engineering*, 2025; 92: 252 DOI: 10.1016/j.ymben.2025.08.011

Conference Report – TRIZCON, Riga



Well, it was nice to get out of the office for a couple of days. Nice to visit a European country I've never been to before, nice to meet some old friends, and nice to make new ones. One or two of the papers being presented weren't bad either. Actually, there were several good ones. Just that most of them had nothing to do with TRIZ. Let's call those ones a horizontal wisdom building opportunity.

The main event was held at the University of Latvia, on the 8th floor of the rather beautiful House of Science on the 6th and 7th of this month. The photo above was taken from the back of the conference room the 25 live attendees sat in for the two days. Again, in fairness, there were often another 8 to 10 people attending online too. So, definitely a step up from the ICSI conference I attended in Turkey last year.

The intertwined theme there also was 'AI'. In my mind there must be millions of people attending AI conferences around the world these days, so, in theory at least, I can picture the attraction of attaching TRIZ to it. In practice, what we're seeing so far are AI people that have no idea what TRIZ is, or any interest in it, and TRIZ people sat in a room watching a presentation in which a TRIZ expert is demonstrating how Chat-GPT (other generative AIs are available) is quite good at summarising what different parts of classical TRIZ are, and terrible at solving problems. I'm not quite sure whether either delivered any meaningful benefit, but I guess, we have to start to find some common ground somewhere.

I was there to present a keynote on innovation timing (slides will be available to download from the 'Free Downloads' page of the SI online shop in due course). Apart from attempting to describe how difficult the timing question is to get right, my main motivation was to try and steer the conversation towards the evolution of TRIZ into the murky world of complex adaptive systems. Classical TRIZ was designed for complicated problems (like chess) and all of the case studies presented during the two days were consistent with that. Inventing a neat solution is complicated. Patenting it is complicated too. But the moment the objective becomes to create something successful, the boundary into complex has been crossed and TRIZ is not going to help any longer. Or rather Classical TRIZ isn't.

I received lots of nice comments after I'd presented, but suspect that the impact in terms of whether anyone was going to do anything different after hearing what I had to say, was

likely close to zero. Which kind of seems to be the case with a lot of conferences that I attend these days. Everyone is on transmit, and no-one is on receive. We're all too busy going around in circles, I guess. Understandable in a lot of domains, but TRIZ – literally founded on the idea of 'someone, somewhere already solved your problem' – really out to be taking a lead here. If it is, it wasn't in Riga.

For anyone interested, here's the agenda for the two conference days:



UNIVERSITY OF
LATVIA

Agenda for TRIZCON2025 event -- Riga, Latvia - 6-7 October 2025



House of Science, University of Latvia, Riga, 8th floor, 801-802 rooms

Monday: 6 October		DAY -1		
Time		Speaker	Country	Description
8:30				Doors open for collecting badges & handouts
9:00		Aigars Atvars	Latvia	Moderator: Greetings and Opening Remarks
		Enno Ence	Latvia	Greeting from the University of Latvia
		Isak Bukhman	USA	Greetings and Introduction to speakers
9:20		Karen Gadd	GB	How TRIZ activates Teams' Human Intelligence
10:00		TRIZ Master Isak Bukhman	USA	Technology for Innovation (TRIZ) Implementation Plan for Companies
10:35	BREAK			
10:50	ZOOM	Dr. Sune Svanberg	Sweden	How to be Inventive in Science - the Case of Laser Spectroscopy
11:30		Darrell Mann	GB	Timing, Timing, Timing: Integrating TRIZ and AI to Navigate the When-To-Innovate Paradox
12:15	LUNCH			
13:15	ZOOM	TRIZ Master, Dr. Pesakh (Pavel) Amnuel	Israel	Scientific and technical forecasting in futurology and science fiction literature
13:55		Dr. Aigars Atvars	Latvia	Advancing Space Research with TRIZ: Problem Solving in Asteroid Identification and Satellite Laser Tracking
14:35		Dr. Anatoly Agulyansky	Israel	Functional Thinking: Invest to Win and Stop Funding Losses
15:15	BREAK			
15:30	ZOOM	TRIZ Master, Dr. Boris Farber	USA	Discovering Smart Adaptive Medical Drugs: The Future of Medicine (Dynamic Self-Adapting Chemical Structures and Their Properties)
16:10	ZOOM	Bryan Cassady	Belgium (Brasil)	TRIZ with AI – Theory and Results
16:50		Aigars Atvars & Isak Bukhman		Conclusion, announcements
18:00				Networking event and dinner
	End of today's Conference			
Tuesday: 7 October		DAY -2		
Time	Category	Speaker	Country	Description
8:30				Doors open for collecting badges & handouts
9:00		Aigars Atvars		Remarks, announcements
9:10		Dr. Aigars Atvars	Latvia	Large Language Model Chatbots for TRIZ: Training and Evaluation of Problem-Solving Capabilities
9:50		TRIZ Master Jūlijs Muraškovskis	Latvia	Laws on the development of systems of scientific ideas
10:30	BREAK			
10:45	ZOOM	Dr. Katarina Svanberg	Sweden	Some challenges needed to be handled in medicine
11:25		Dr. Fatima Ezzahra Essaber	Morocco	Toward Inventive Sustainable Supply Chains: TRIZ in Practice and AI in Prospect for Lean Green Systems
12:00	LUNCH			
13:00		Dr. Yuli Chakk	Israel	TRIZ and the Complementarity Principle in Science and Technology
13:40		Dr. Ingrida Murashkovska	Latvia	Investigation of Adults' Lifelong Learning – an Example of Using TRIZ in Social Research
14:20		TRIZ Master Alexander Theodor Narbut	Ukraine	Telescope of G.S. Altshuller
14:50	BREAK			
15:05	ZOOM	Dr. Li Huangye & Jason Wang	China	AI - enabled innovative educational practice activities in universities
15:45		Dr. Temesgen Getachew Gatilaw	Ethiopia	Problem Sensemaking Under Deep Uncertainty: Speed, Precision, and Market Relevance (SPM)-inspired TRIZ problem formulation
16:25		Laili Sakijeva	Latvia	TRIZ and the evolution of language
16:55		Aigars Atvars & Isak Bukhman		Conclusion
17:00	End of Conference -- Thank you!			

Meanwhile, back at my hotel – the once regal Radisson Blu hotel at Daugava – it was inspiring to see that I'd been booked into the same hotel as a veritable armada of world

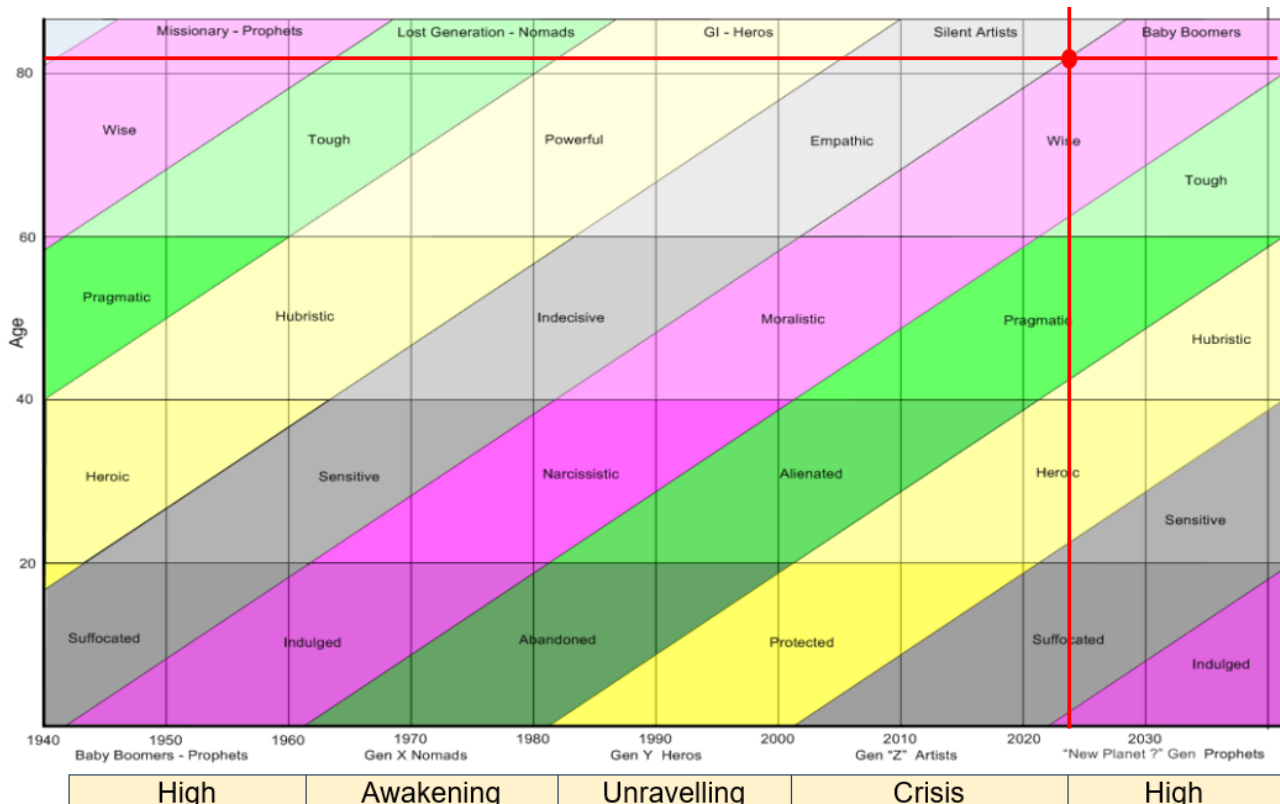
famous dignitaries and celebrities. I'm not sure whether The Queen or George W. Bush ever had to worry about making themselves a cup of tea in the morning, but if they did, I wonder if they possessed the requisite TRIZ knowledge to work out how to plug the kettle in. Here was my rather precarious Principle 12 solution...



...told you I learned something.

Generational Cycles – From Duty to Self: How Generational Shifts in Wealth Are Redefining Legacy and Philanthropy

The average life expectancy for UK citizens is currently just over 81 years (79 years for men; 83 years for women). This is significant because it means this year sees the average 81 right at the boundary between the Silent and Baby Boomer generation:



These transition points are always important. They mean a step-change in the ways society functions. Particularly, in this case in the ways people think about retirement and inheritance.

Both are among the most revealing indicators of how societies think about continuity, duty, and value. As the Silent Generation and the Baby Boomers enter their elder years, their contrasting attitudes toward wealth, legacy, and philanthropy are shaping one of the largest wealth transfers in human history. Understanding these differences is not only useful for families but essential for charities and foundations that hope to remain relevant.

The Silent Generation: Duty and Legacy

Born between the mid-1920s and early 1940s, the Silent Generation grew up in the shadow of the Great Depression and the upheaval of the Second World War. As children, scarcity and insecurity were formative experiences. As adults, they became savers, builders, and joiners – committed to duty, loyalty, and quiet stability.

For many Silents, the concept of legacy is inseparable from responsibility. To die without leaving something behind feels like a dereliction of duty. To family, to community, and to society at large. Their philanthropy often reflects this ethic: they endow scholarships, donate to churches and universities, and build funds that ensure their name continues in association with stability and generosity.

Charities have long understood how to appeal to this ethos. Campaigns emphasising permanence, gratitude, and continuity resonate powerfully with Silents. They want assurance that their gift will be carefully stewarded, and they often prefer vehicles such as bequests, endowments, and foundations that promise to outlast them. For this generation, giving is an extension of the same thrift and responsibility that shaped their adult lives.

The Baby Boomers: Self and Experience

The Baby Boomers, born between the mid-1940s and early 1960s, came of age in an era of postwar 'High', cultural upheaval, and individual liberation. They are the generation that embraced "self-expression," "personal growth," and "living for today." They're also, as the name suggests, a big generation – around 13.6M people, as opposed to around 4.7M members of the Silent generation.

For many Boomers, retirement is less about preserving wealth than about consuming it meaningfully. Popular culture captured this in the tongue-in-cheek acronym "SKI", Spending the Kids' Inheritance. Travel, lifestyle upgrades, wellness investments, and second homes have become central features of their retirement years.



This does not mean that Boomers lack generosity. Rather, their giving is often more experiential and immediate. They are drawn to living legacies: donations that allow them to witness their impact firsthand. Unlike the Silent Generation, which often looked to posterity, Boomers prefer to see and feel the difference their contribution makes today.

Yet this narrative of consumption is now complicated by a dawning awareness. Many Boomers look at their children and grandchildren facing precarious housing markets, insecure pensions, and mounting climate anxieties, and feel a pang of guilt. Having benefited from rising asset prices and relatively stable careers, Boomers increasingly recognise that the generational ladder has been pulled up behind them. This is prompting some to re-examine their attitudes toward legacy.

Duty vs. Freedom, Family vs. Community

The contrast between these two generations can be summed up in a few tensions:

Duty vs. Freedom: Silents often feel guilty if they don't leave a legacy; Boomers often feel guilty if they do, as if it signals a life not fully lived.

Family vs. Community: Silents frequently divide their legacies between family and institutions. Boomers lean more heavily toward family, yet are now re-balancing as they witness societal challenges.

Accumulation vs. Experience: Silents accumulated wealth to pass on; Boomers prioritised spending and experiences, sometimes at the expense of inheritance. Companies like McCarthy & Stone have already begin to exploit these differences in the way they have re-invented 'care homes' for the elderly. Peter-Pan-like Boomers won't live in a Care Home, but they will live in places that emphasise their continuing journey of life-long learning ('Scholar's Place')... and McCarthy & Stone are making sure they take their slice of the inheritance for their efforts.

Implications for Charities

For charities and non-profits, these differences require tailored strategies.

Reaching the Silent Generation:

- Emphasise duty, gratitude, and tradition.
- Highlight permanence: "Your gift will educate students in perpetuity."
- Provide confidence in stewardship through clear governance and strong institutional reputation.

Reaching the Boomers:

- Appeal to self-actualisation: "See the lives you transform while you are alive."
- Offer flexible giving models: living legacies, donor-advised funds, social-impact investments.
- Address guilt with messaging that balances enjoyment of wealth with societal contribution: "Share your success without giving up your freedom."

Both generations want to make a difference, but they are motivated by profoundly different stories: one rooted in duty and permanence, the other in freedom and lived experience.

Looking Ahead

The Silent Generation is rapidly fading, and their sense of duty-driven philanthropy will fade with them. As the Boomers pass through their final decades, charities will encounter donors who want to participate actively in their giving, who expect transparency, and who want to align philanthropy with their personal narrative of self-fulfilment.

This transition means that traditional estate-based giving will decline in relative importance, replaced by more immediate, participatory forms of philanthropy. The organisations that adapt – by recognising the deep generational psychology behind giving – will be best positioned to benefit from what is projected to be the largest inter-generational wealth transfer in history.

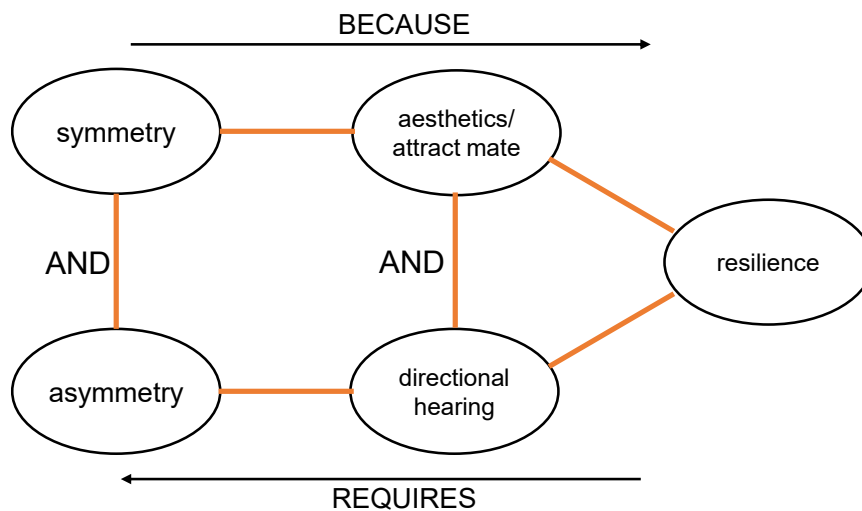
Retirement and inheritance are no longer simply about savings accounts and wills. They have become reflections of how different generations interpret meaning, responsibility, and selfhood. The Silent Generation sought to leave the world more stable than they found it. The Boomers seek to live fully while they can, but may yet leave a legacy if charities can help them reconcile freedom with responsibility.

Biology – Human Ear

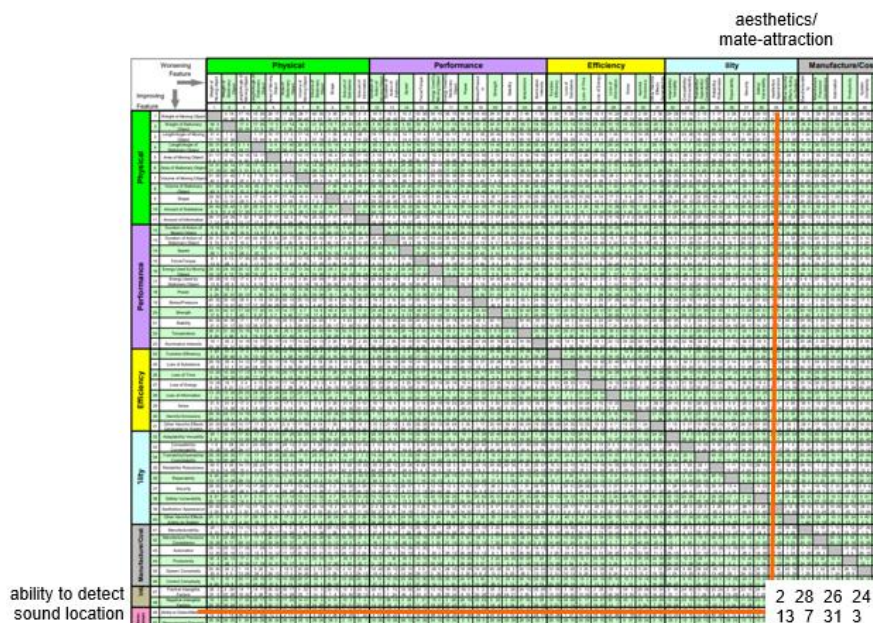


Strange that we've never really featured humans in this section of the ezine before. We're biology too, and share a lot of evolutionary traits and directions with other cousins in Nature. Owls are a fairly distant cousin, but we share with them what we share with other life-forms: a symmetry related contradiction.

On the one hand, symmetrical features – especially facially – are generally speaking judged to be aesthetically pleasing, and therefore a helpful resource when it comes to finding a mate. On the other hand, asymmetry tends to come attached to all sorts of functional benefits. Most notably in the human/owl case, the ability to hear where noises are coming from. An important skill for detecting both predators and potential prey. Here's what this contradiction looks like mapped onto a Bubble Map:



Here's what we find when we map the middle-bubble-pair, aesthetics-versus-directional hearing, onto the Contradiction Matrix:



Interesting to see how the owl solves the contradiction differently to the human – it uses Principle 24, Intermediary and covers its ears with a symmetrical arrangement of feathers. Probably helpful, too, to ensure that aerodynamic forces are balanced when the owl is flying.



Humans tend to do less flying, unless it's inside an Airbus, so aerodynamic drag is less of an issue. We've evolved a solution that, per the photo at the head of the article looks beautiful when looking directly into a person's face (the most important mate-attraction angle, I'm guessing), and is small enough or easily enough to cover with our own version of feathers when viewed from the side:



So much for the aesthetic part of the human contradiction. How do our ears help with detecting the whereabouts of the sound we hear?

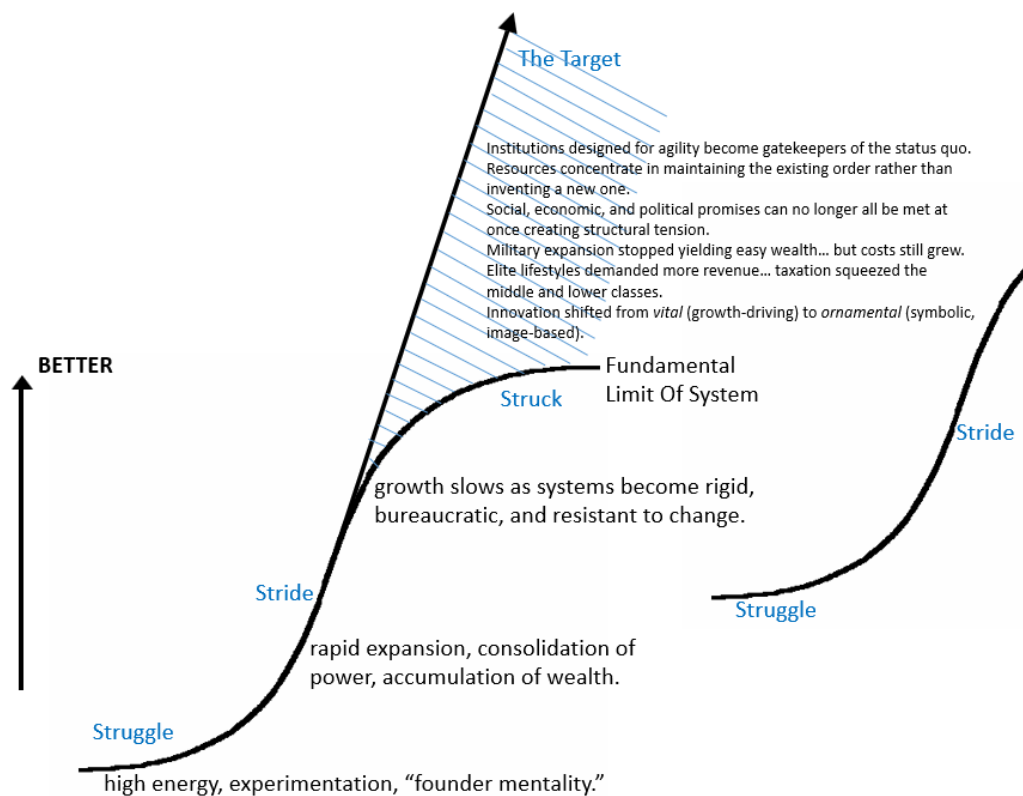
Answer a combination of Inventive Principles 3, Local Quality, and 7, Nested Doll, in the form of the array of near concentric ribs and folds.

By way of a lovely demonstration of how we're unable to detect sound location if we didn't have these geometric features, there's a video of a lovely experiment by Mark Rober (<https://www.youtube.com/shorts/eW8ZNkuLhZw>) where he fills the ribs and folds with Play-Doh to make them smooth. Then asks a blindfolded, smooth-eared subject to try and work out where sounds are coming from. It's quite funny how clueless we become.



Principles 3 and 7 work directional miracles for us, giving us the *local* asymmetry we need to detect sound, and the *global* symmetry we need to help attract a mate.

Short Thort



Oftentimes it is at the steepest slope of the S-curve that enterprises set their future growth targets – ‘keep on this trajectory’ everyone is told. Thus setting in train all the accidental harm that comes from trying to go beyond the fundamental limits of the system. The further into the hatched area an enterprise is allowed to travel, the greater the harm – internal and external – it ends up causing. All the time covered by the Unaccountability Machine. The Machine that enables everyone to say, when the harms inevitably come to light, ‘this wasn’t my fault’.

News

SI LMS

Starting this month, we’ve finally got around to making our 30hour SI learning management system content available to individual subscribers. Originally targeted at corporate users seeking a TRIZ/SI learning programme chunked into lunchbreak sized sections, we thought we’d explore whether the same idea might make sense for individual subscribers. We’re making the 60 sessions available in two forms: the simplest allows subscribers three-months’ worth of access to the LMS, allowing them to work through the content at their own pace. The second option allows subscribers to take a formal evaluation at the end of their journey with the LMS and, assuming that evaluation is passed, receive our Level 1 Certificate. More details on both options in the SI online shop.

TRIZ Mastery Hub

Following a third session for the TRIZ Mastery Hub this month, Darrell has been invited back to do a fourth on 15 December. Probably not surprisingly, the focus this time around will be 1%ers. More details at the Mastery Hub website:

<https://triz-mastery-hub.circle.so/c/special-events/darrell-mann-the-1-er>

1%er Book

If you've read about our forthcoming 1%er book in either of the articles in this month's ezine, you may be pleased to know that the ebook version will be receiving a 'soft launch' before the end of this month. Most likely via the SI online shop, but watch this space in case we decide to take a different track. Like (deep breath) Amazon. Expect to read more on the strategy next month, including dates for the next round of 1%er online workshops.

European Seminar on AI for Sustainability and Climate Change

Well, this one's a nice surprise. We thought there'd be millions of people submitting papers for this online conference, scheduled to happen on 15 January next year, but somehow, we've ended up as one of the lucky ones selected to present on the day. Our subject: 'Beyond Trade-offs: How Multi-Agent AI Will Unlock the Next Generation of SDG Breakthroughs'. More details of the conference here: <https://www.haw-hamburg.de/detail/news/news/show/european-seminar-on-ai-for-sustainability-and-climate-change/>

New Projects

This month's new projects from around the Network:

- Education – Requisite AI-agent App Development Project
- NGO – Innovation Strategy Project
- Medical Devices – Innovation Strategy Workshops
- Electronics – Innovation Leadership Programme Design
- Advertising – 1%er Assessment Project
- Finance – 1%er Dashboards
- Consulting – Business Strategy Project
- Communication – SI LMS

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